Limits and Continuity

We say $\lim_{x \to a} f(x) = L$ if $f(x)$ close enough to L for any x close enough to a

Limit Laws

$L, M, c, k \in R$

$\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$ then

$\lim_{x \to c} (f(x) + g(x)) = L + M$

$\lim_{x \to c} (f(x) - g(x)) = L - M$

$\lim_{x \to c} (k \cdot f(x)) = k \cdot L$

$\lim_{x \to c} (f(x) \cdot g(x)) = L \cdot M$

$\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{L}{M}$ $M \neq 0$

$\lim_{x \to c} [f(x)]^n = L^n$ n a positive integer

$\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{L}$ n a positive integer

If $P(x) = a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \ldots + a_0$ then $\lim_{x \to c} P(x) = P(c)$

$$\lim_{x \to c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}$$

The Sandwich Theorem

Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c, except possibly at $x = c$ itself. Suppose also that

$$\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L \text{ then } \lim_{x \to c} f(x) = L$$

$-|\theta| \leq \sin\theta \leq |\theta|$

$-|\theta| \leq 1 - \cos\theta \leq |\theta|$

14/10/2022

Precise Definition of Limit

$\lim_{x \to a} f(x) = L \iff$ for any given $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < |x-a| < \delta$ then $|f(x)-L| < \varepsilon$

$\lim_{x \to 0} \frac{\sin x}{x} = 1$ (x in radias)

Continuous Functions

If function is continious at $x=a$ $\lim_{x \to a} f(x) = f(a)$

- Continuity test

1) $f(c)$ exists 2) $\lim_{x \to c} f(x)$ exists 3) $\lim_{x \to c} f(x) = f(c)$

- Properties of Continuous Functions

If the functions f and g are continuous at $x=c$ then the following algebraic combinations are continuous at $x=c$

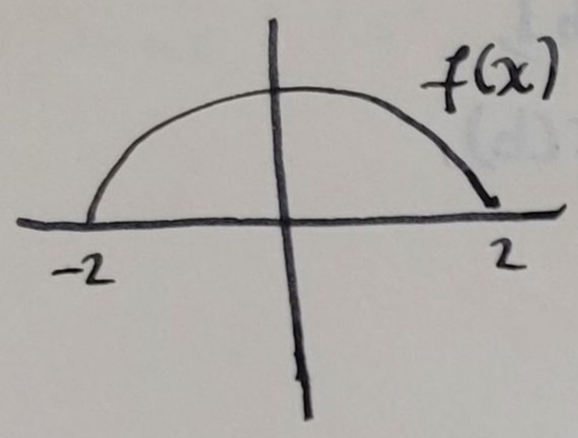
- $f+g$
- $f-g$
- $k \cdot f$ (for any number k)
- $f \cdot g$
- f/g ~~$g \neq 0$~~ $g(c) \neq 0$
- f^n (n a positive integer)
- $\sqrt[n]{f}$ provided it is defined on an interval containing c, where n is a positive integer
- Every polinomial function is cts.

- If f is continuous at c and g is continuous at $f(c)$, then the composition $g \circ f$ is continuous at c.
- If $\lim_{x \to c} f(x) = b$ and g is continuous at the point b, then

$$\lim_{x \to c} g(f(x)) = g(b) = g\left(\lim_{x \to c} f(x)\right)$$

- $f(x) = |x|$ is cts. at every $x \in R$

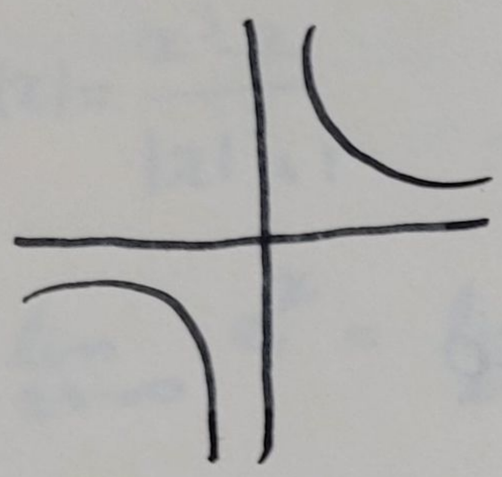
$f(x)=\sqrt{4-x^2}$, $[-2,2]$



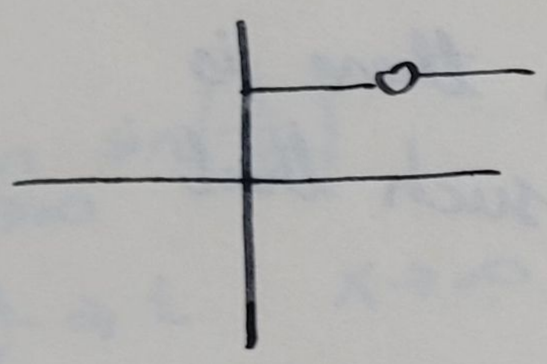
We say f is cts. on a closed interval $[a,b]$ if it is right cts. at a and left cts. at b for $a<x<b$

f is right cts. at $x=-2$
f is left cts. at $x=+2$
f is cts. at $-2<x<2$

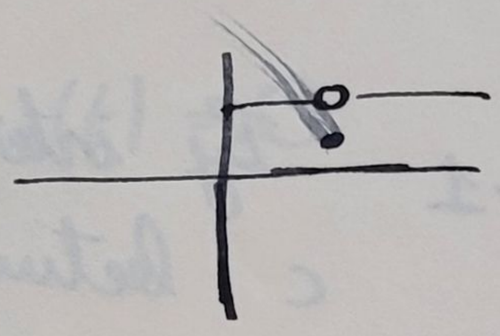
} So f is cts. on $[-2,2]$



$y=\frac{1}{x}$ is cts. functions is everywhere because x can not be equal to zero



cts. functions



not cts. functions

15/10/2019

-Continuity of trigonometric functions

$f(x)=\sin x$ cts. at everywhere $x\in R$

$\lim_{x\to c}\sin x=\sin c$ $\quad x\to c \Leftrightarrow h\to 0$ $\quad \lim_{h\to 0}\sin(c+h)=\sin c$

$\lim_{h\to 0}\sin(c+h)=\lim_{h\to 0}\sin c\cdot\cos h+\sin h\cdot\cos c$

$\lim \sin c\cdot\cos h+\lim_{h\to 0}\sin h\cdot\cos c \quad \sin c\cdot\lim_{h\to 0}\cos h+\cos c\cdot\lim_{h\to 0}\sin h$

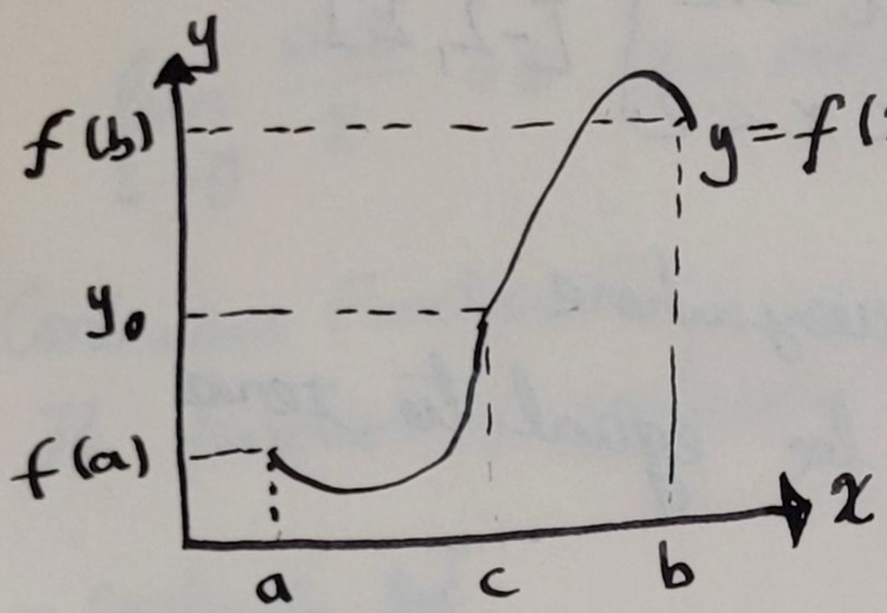
$\sin c\cdot 1+\cos c\cdot 0=\sin c$

$\tan x=\frac{\sin x}{\cos x}$ is cts. wherever $\cos x\neq 0$ since $\sin x$ and $\cos x$ cts.

$\cot x$, $\sec x$, $\operatorname{cosec} x$ are also cts. wherever they are defined

The Intermediate Theorem

If f is a cts. function on a closed interval $[a,b]$, and if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a,b]$



Show that there is a root of the equation $x^3 - x - 1 = 0$ between 1 and 2

$f(x) = x^3 - x - 1$

$f(1) = -1$

$f(2) = 5$

$f(3) = 23$

By intermediate theorem there is c between 1 and 2 such that

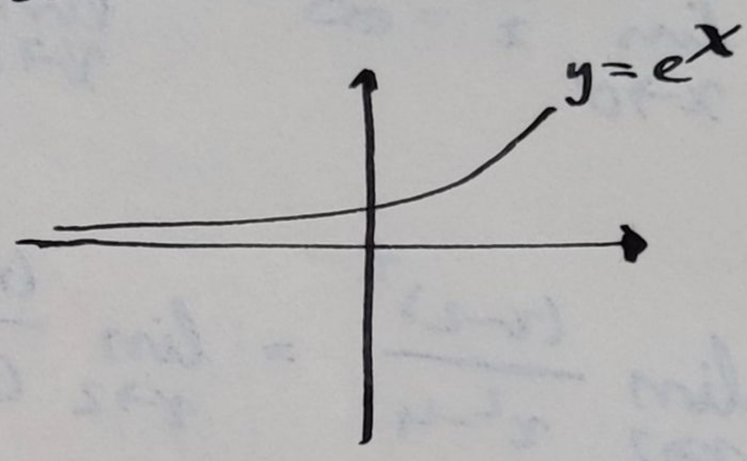
$f(c) = 0$

21/10/2019

- Find the horizontal asyptotes of

$$f(x)=\frac{x^3-2}{|x|^3+1}$$

$$\lim_{x\to\infty} f(x)=\frac{x^3+2}{x^3+1}=1 \qquad \lim_{x\to-\infty}\frac{x^3-2}{-x^3+1}=-1$$



- $\lim_{x\to-\infty} e^x = \lim_{x\to\infty} e^{-x} = \lim_{x\to\infty}\frac{1}{e^x}=0$

- $\lim_{x\to\infty}\sin\left(\frac{1}{x}\right)$ $\qquad \lim_{t\to0^+}\sin(t)=0$

$\frac{1}{x}\Rightarrow t \qquad x\to\infty \Rightarrow t\to0^+$

- $\lim x\cdot\sin\left(\frac{1}{x}\right) \qquad x\Rightarrow\frac{1}{t} \qquad \lim_{t\to0^+}\left(\frac{1}{t}\right)\cdot\sin t=1$

$x\to\pm\infty$

- $\lim_{x\to0^-} e^{\frac{1}{x}} \qquad x\Rightarrow\frac{1}{t} \qquad \lim_{t\to-\infty} e^t=0$

$x\to0^- \Rightarrow t\to-\infty$

- Find horizontal asymptotes of $y=2+\frac{\sin x}{x}$ $\qquad \lim_{x\to\infty} 2+\frac{\sin x}{x}$

- $\lim_{x\to\infty}\left(x-\sqrt{x^2+16}\right)=\lim_{x\to\infty}\frac{(x-\sqrt{x^2+16})(x+\sqrt{x^2+16})}{(x+\sqrt{x^2+16})}=0$

Oblique Asymptote

$$f(x) = \frac{x^2-3}{2x-4}$$

$$\begin{array}{r|l} x^2-3 & 2x-4 \\ \hline -x^2+2x & \frac{1}{2}x+1 \\ \hline 2x-3 & \\ -2x+4 & \\ \hline 1 & \end{array}$$

$$\frac{x^2-3}{2x-4} = \frac{1}{2}x^2+1+\frac{1}{2x-4}$$

$$\lim_{x\to\infty} \frac{x^2-3}{2x-4} = \lim_{x\to\infty} \frac{1}{2}x+1$$

Infinite limits

$$\lim_{x\to 0^+} \frac{1}{x} = \infty \qquad \lim_{x\to 0^-} \frac{1}{x} = -\infty$$

} Means does not exists because the function grows out of bounds as $x \to 0^+$

$$\lim_{x\to 2} \frac{(x-2)^2}{x^2-4} = \lim_{x\to 2} \frac{(x-2)(x-2)}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x-2}{x+2} = 0$$

$$\lim_{x\to 2} \frac{x-3}{x^2-4} \text{ does not exists}$$

$$\lim_{x\to 2^+} \frac{(x-3)}{(x+2)(x-2)} = -\infty$$

$$\lim_{x\to 2^-} \frac{(x-3)}{(x+2)(x-2)} = +\infty$$

$$\lim_{x\to\infty} \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots}{b_m x^m + b_{m-1}x^{m-1} + \cdots} = \begin{cases} +\infty & \text{if } n>m \text{ and } \frac{a_n}{b_m} > 0 \\ -\infty & \text{if } n>m \text{ and } \frac{a_n}{b_m} < 0 \end{cases}$$

Vertical asyptote

A line $x=a$ is a vertical asymptote of the graph of a function $y=f(x)$ if either

$$\lim_{x\to a^+} f(x) = \pm\infty \qquad \lim_{x\to a^-} f(x) = \pm\infty$$

$y = \frac{-x}{x^2-4}$ vertical asy. $-2, 2$

$\lim_{x\to 0^+} \ln x = -\infty$

04/11/2019

2.3 Differtiation Rules

- $(fg)' = f' \cdot g + g' \cdot f$
- $(f_1 f_2 f_3)'$
- Theorem: If f is differentiable and $f(x) \neq 0$ then $\frac{d}{dx} \cdot \frac{1}{f(x)} = \frac{-f'(x)}{f(x)^2}$ or $\left(\frac{1}{f}\right)' = \frac{-f'}{f^2}$

Proof

$$\frac{d}{dx}\left(\frac{1}{f(x)}\right) = \lim_{h \to 0} \frac{\frac{1}{f(x+h)} - \frac{1}{f(x)}}{h}$$

$$= \lim_{h \to 0} \frac{f(x) - f(x+h)}{h\, f(x+h) f(x)}$$

$$= \left[\lim_{h \to 0} \frac{f(x) - f(x+h)}{h}\right]\left[\lim_{h \to 0} \frac{1}{f(x+h) f(x)}\right]$$

$$= -\left[\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}\right]\left[\lim_{h \to 0} \frac{1}{f(x+h) f(x)}\right]$$

$$= -f'(x) \cdot \frac{1}{f(x)^2}$$

- If f, g are differentiable at x and $g(x) \neq 0$ then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x) g(x) - f(x) g'(x)}{g(x)^2}$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Proof

$$\left(\frac{f}{g}\right)' = \left(f \cdot \frac{1}{g}\right)' = f' \cdot \frac{1}{g} + f\left(\frac{1}{g}\right)'$$

$$= \frac{f'}{g} + f\left(\frac{-1}{g^2} \cdot g'\right)$$

Power Rule: $\frac{d}{dx} x^r = r x^{r-1}$ (proved this positive integers)

ex. $y = \frac{x^5}{x^{\frac{2}{3}}+1}$, $y' = ?$ $\quad y' = \frac{5x^4(x^{\frac{2}{3}}+1) - x^5(\frac{2}{3}x^{-\frac{1}{3}}+0)}{(x^{2/3}+1)^2}$ (minus)

perpendicular line: değil

ex. Find the x-coordinate on the curve $y = \frac{x+1}{x+2}$ where the tangent line is parallel to the line $y = 4x$

slope $= 4$

$y' = 4$ for x

$y' = \frac{(x+2)-(x+1)}{(x+2)^2}$

$x = \{\frac{-3}{2}, \frac{-5}{2}\}$

ex $f(2) = 2$, $f'(2) = 3$ $\quad \frac{d}{dx}\left(\frac{x^2}{f(x)}\right)\Big|_{x=2}$

$\frac{2xf(x) - f'(x)\cdot x^2}{f^2(x)}$ (-1)

3.5 Derivatives of Trigonometric Functions

$\frac{d}{dx}\sin x = \cos x$

Proof:

$\frac{d}{dx}\sin x = \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$

$= \lim_{h\to 0} \frac{\sin x\cos h + \sin x\cos x - \sin x}{h}$

$= \lim_{h\to 0} \sin x\left(\frac{\cos h - 1}{h}\right) + \lim_{h\to 0} \cos x \frac{\sin h}{h}$

$= \sin x \underbrace{\lim_{h\to 0} \frac{\cos h - 1}{h}}_{0} + \cos x \underbrace{\lim_{h\to 0} \frac{\sin h}{h}}_{1} = \cos x$

ex

Show $\lim_{h\to 0} \frac{\cosh-1}{h} \cdot \frac{\cosh+1}{\cosh+1} = \lim_{h\to 0} \frac{\cos^2 h - 1}{h(\cosh+1)} = \lim_{h\to 0} \frac{-\sin^2 h}{h(\cosh+1)}$

$= \underbrace{-\lim_{h\to 0} \frac{\sinh}{h}}_{-1} \underbrace{\lim_{h\to 0} \frac{\sinh}{\cosh+1}}_{\frac{0}{1}} = 0$

HW

$\frac{d}{dx} \cot x = -\csc^2 x$

$\frac{d}{dx} \sec x = \sec x \tan x$

$\frac{d}{dx} \csc x = -\csc x \cdot \cot x$

3.6 Chain Rule

$y = f \circ g(x) = f(g(x))$

$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$

if $y = f(x)$, $u = g(x)$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

ex $y = \sqrt{x^2+1}$ $\frac{dy}{dx} = ?$ $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$u = x^2+1$

$y = u^{\frac{1}{2}}$

$= 2x \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{u}} = \frac{2x}{2\sqrt{x^2+1}}$

$y=(x^3+1)^{1000}$ $\quad u=x^3+1$

$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}=1000u^{999}(3x^2)$

ex $\frac{d}{dx}\left(x^2\cdot\cos(\sqrt{x})\right)=2x\cos(\sqrt{x})+x^2\cdot\left(-\sin(\sqrt{x})\cdot\frac{1}{2\sqrt{x}}\right)$

$\sqrt{x}=u$ $\quad u^4\cdot\cos(u)$ $\quad \frac{dy}{du}\cdot\frac{du}{dx}=\frac{1}{2\sqrt{x}}\cdot\left(4u^3\cdot\cos(u)+\sin(u)\cdot u^4\right)$

$x^2=u^4$

ex $\frac{d}{dx}|x|=\frac{d}{dx}\left(\sqrt{x^2}\right)=\frac{1}{2\sqrt{x^2}}\cdot 2x=\frac{x}{|x|}$, $x\neq 0$

ex $\frac{d}{dx}|\tan(3x)|=\frac{\tan(3x)}{|\tan 3x|}\cdot\sec^2 3x\cdot 3$

ex. $y=\sin\left(\tan x+\sqrt{2x+1}\right)^3$

$\frac{dy}{dx}=?$ $\quad 3\sin\left(\tan x+\sqrt{2x+1}\right)^2\cdot\cos\left(\tan x+\sqrt{2x+1}\right)^3\cdot\left(\sec^2 x+\frac{1}{2\sqrt{2x+1}}\cdot 2\right)$

05/11/2019

Derivatives of Exponential Functions

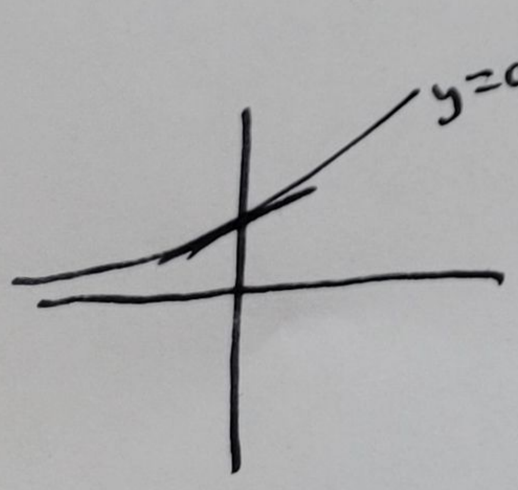
$y=a^x$, $\quad a>0$, $\quad x\in\mathbb{R}$

$y'=\lim_{h\to 0}\frac{a^{x+h}-a^x}{h}$ $\qquad L=\lim_{h\to 0}\frac{a^h-1}{h}$

$=\lim_{h\to 0}a^x\left(\frac{a^h-1}{h}\right)$ $\qquad \frac{d}{dx}a^x=L\cdot a^x$

$=a^x\left[\lim_{h\to 0}\frac{a^h-1}{h}\right]$ $\qquad L=\lim_{h\to 0}\frac{a^h-a^0}{h}=\frac{d}{dx}a^x\Big|_{x=0}$

slope of the tangent line at $x=0$ to $y=a^x$



If $a=2\Rightarrow L\approx 0.69$

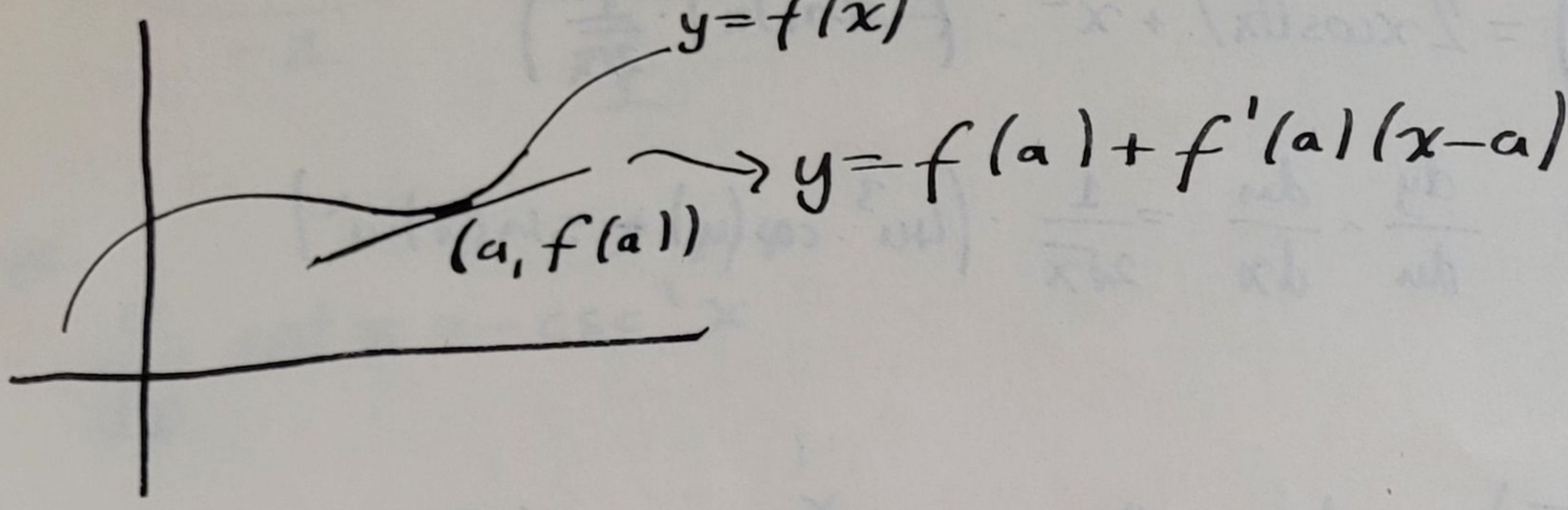
$a=2.5\Rightarrow L\approx 0.92$

$a=3\Rightarrow L\approx 1.1$

$a=e\Rightarrow L=1$

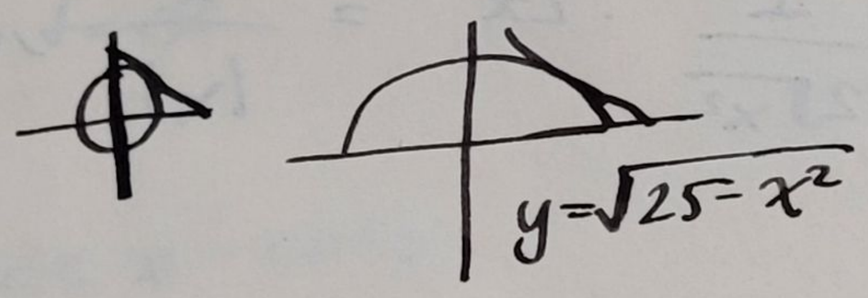
$$\frac{d}{dx} e^{\tan x^2} = e^{\tan x^2} \cdot \sec^2 x^2 \cdot 2x$$

3.7 Implicit Differentation



$$\rightarrow y = f(a) + f'(a)(x-a)$$

ex. What is the tangent line to curve $x^2+y^2=25$ at the point (3,4)?



1st way

$$\frac{dy}{dx} = \frac{-x}{\sqrt{25-x^2}}\Big|_{x=3} = -3/4$$

2nd way

Near (3,4) $y = y(x)$

$$x^2 + y(x)^2 = 25$$

$$\frac{d}{dx}\left(x^2 + y(x)^2\right) = \frac{d}{dx} 25 = 0$$

$$2x + 2y(x) \cdot y'(x) = 0$$

$$y'(3) = \frac{-3}{4}$$

ex Find an equation of the tangent line to the curve $x \cdot \sin(xy - y^2) = 0$ at (1,1)

$$1 \cdot \sin(xy-y^2) + x \cdot (\cos(xy-y^2)) \cdot (1 \cdot y + x \cdot y' - 2yy') = 0$$

$x=1 \quad y=1$

$$\sin(1\cdot 1 - 1) \to 0 \qquad \cos(1\cdot 1 - 1) \to 1$$

$$1 \cdot (1 \cdot 1 + 1 \cdot y' - 2 \cdot 1 \cdot y') = 0$$

$$1 + y' - 2y' = 0$$

$$y' = 1 = \text{slope}$$

$$y - 1 = 1(x-1)$$

- $y^2 = e^{x^2} + 2x$ Find $\frac{dy}{dx}$ in terms of x, y

$$\frac{d}{dx} y^2 = \frac{d}{dx}(e^{x^2} + 2x) \Rightarrow 2 \cdot y \cdot y'(x) = e^{x^2} \cdot 2x + 2$$

$$y'(x) = \frac{e^{x^2} \cdot 2x + 2}{2y}$$

- $y^2 = e^{x^2} + 2x$

Find $\frac{d^2y}{dx^2}$ in terms of x, y, y'

$$y'' = \frac{(e^{x^2} \cdot 2x \cdot 2x + e^{x^2} \cdot 2 + 0) \cdot 2y - (e^{x^2} \cdot 2x + 2) \cdot 2y'}{(2y)^2}$$

- Prove the power law

$$\frac{d}{dx} x^{m/n} = \frac{m}{n} \cdot x^{\frac{m}{n} - 1}, \quad m, n \in \mathbb{Z}$$

$y = x^{m/n}$ $\quad y^n = x^m$ $\quad \frac{d}{dx} y^n = \frac{d}{dx} x^m = m \cdot x^{m-1}$

$$n \cdot y^{n-1} \cdot y' = m \cdot x^{m-1}$$

$$y' = \frac{m}{n} \cdot \frac{x^{m-1}}{y^{n-1}}$$

$$y^{n-1} = \frac{y^n}{y} = \frac{x^m}{x^{m/n}} = x^{m - m/n}$$

$$y' = \frac{m}{n} \cdot \frac{x^{m-1}}{x^{m - m/n}} = \frac{m}{n} \cdot x^{\frac{m}{n} - 1}$$

3.8 Derivatives of Inverse Functions and Logarithms

$f(x) = 3x+1$ $\quad f^{-1}(x) = \frac{x-1}{3}$

Find $\frac{d}{dx} f^{-1}(x)$ $\quad \frac{d}{dx} f^{-1}(x) = \frac{1}{3}$

2nd way

$f^{-1}(f(x)) = x$ $\quad \frac{d}{dx} f^{-1}(3x+1) = \frac{d}{dx} x = 1$

$f^{-1}(3x+1) = x$

$\left(\frac{d}{dx} f^{-1}(3x+1)\right) \cdot 3 = 1$ $\quad \frac{df^{-1}(3x+1)}{1} = \frac{1}{3}$

When is $3x+1 = 1$ $\quad x = 0$

$\left(\frac{d}{dx} f^{-1}\right)(1) = 1/3$

General case

$f^{-1}(f(x)) = x$

$y = f(x)$

$\frac{df^{-1}}{dx}\Big|_y \cdot \frac{dy}{dx} = 1$

$\frac{df^{-1}}{dx}\Big|_y \cdot \frac{dy}{dx} = 1$

$\frac{df^{-1}}{dx}\Big|_y = \frac{1}{f'(x)}$

$\frac{d}{dx} f^{-1}(y) = x \quad \frac{d}{dx} = 1$

$$\boxed{\frac{d}{dx} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}}$$

$$\frac{df^{-1}}{dx}(f(x)) = \frac{1}{f'(x)}$$

ex: Let $f(x)=x^3-2, \quad x>0$

Find $\frac{df^{-1}}{dx}$ at $x=6$

$f^{-1}(f(x))=x$

$f^{-1}(x^3-2)=x$

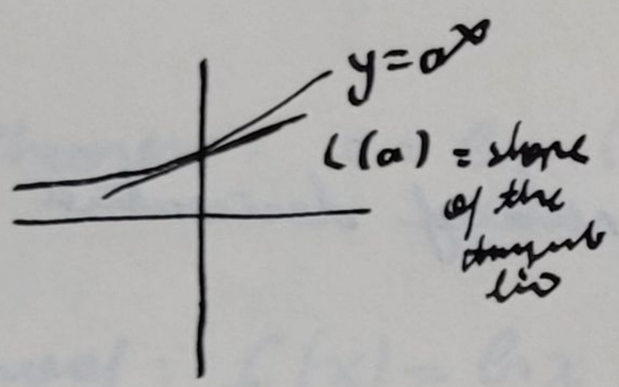
$\frac{d}{dx} f^{-1}(x^3-2)=1$

$\frac{d}{dx} f^{-1}\Big|_{x^3-2} \cdot 3x^2=1$

$\left.\frac{df^{-1}}{dx}\right|_{x^3-2} = \frac{1}{3x^2}$ When is $x^3-2=6$ $x=2$ $\left.\frac{df^{-1}}{dx}\right|_6 = \frac{1}{12}$

18/11/2019

$\frac{d}{dx}a^x = \lim_{h\to 0}\frac{a^{x+h}-a^x}{h} = \lim_{h\to 0} a^x\cdot\frac{a^h-1}{h} = a^x\cdot\lim_{h\to 0}\frac{a^h-1}{h} = L(a)\cdot a^x$



$a=e\approx 2.71 \Rightarrow L(e)=1 \quad \frac{d}{dx}e^x = e^x$

$e^x=y \Leftrightarrow x=\ln y, \quad y>0, \quad x\in\mathbb{R}$

$\frac{d}{dx}\ln x$: $\quad x=e^{\ln x} \Rightarrow \frac{d}{dx}x = \frac{d}{dx}e^{\ln x} \Rightarrow 1 = e^{\ln x}\cdot\frac{d}{dx}\ln x$ $\boxed{\frac{1}{x}=\frac{d}{dx}\ln x}$ $x>0$

$\frac{d}{dx}\ln(f(x)) = \frac{1}{f(x)}\cdot f'(x), \quad f(x)>0$

ex $\frac{d}{dx}\ln(x^2+3) = \frac{1}{x^2+3}\cdot 2x$

ex $\frac{d}{dx}a^x = L(a)\cdot a^x = \frac{d}{dx}e^{\ln a^x} = \frac{d}{dx}e^{x\ln a} = e^{x\ln a}\cdot\ln a$

$= a^x\cdot\ln a$

$a>0$, $x\in\mathbb{R}$ $\boxed{\frac{d}{dx}a^x = a^x\cdot\ln a}$ $\quad \frac{d}{dx}a^{f(x)} = a^{f(x)}\cdot\ln a\cdot f'(x)$

$\frac{d}{dx}\log_a x$: ① $\frac{d}{dx}\frac{\ln x}{\ln a} = \frac{1}{x\cdot \ln a}$

② ~~$\log_a a^x = x$~~ ~~$\frac{d}{dx}x = \frac{d}{dx}\log_a a^x$~~ $\frac{d}{dx}x = \frac{d}{dx}a^{\log_a x}$

$x = a^{\log_a x}$, $x>0$

~~$1 =$~~ $1 = a^{\log_a x}\cdot \ln a \cdot \frac{d}{dx}\log_a x$

$\frac{d}{dx}\log_a x = \frac{1}{\ln a \cdot a^{\log_a x}}$

$= \frac{1}{x\cdot \ln a}$

ex

$\frac{d}{dx}3^{\sin x} = 3^{\sin x}\cdot \ln 3 \cdot \cos x$

ex

$y = \frac{(x^2+1)(x+3)^{1/2}}{x-1}$, $x>1$ 1st way: Usual rule of derivative

2nd way: $\ln y = \ln\left[\frac{(x^2+1)(x+3)^{1/2}}{x-1}\right]$

$\ln y = \ln(x^2+1) + \ln(x+3)\cdot\frac{1}{2} - \ln(x-1)$

$\frac{d}{dx}\ln y = \frac{1}{y}\cdot\frac{dy}{dx} = \frac{1}{x^2+1}\cdot 2x + \frac{1}{2}\cdot\frac{1}{x+3} - \frac{1}{x-1}$

$\frac{dy}{dx} = \left[\frac{(x^2+1)(x+3)^{1/2}}{x-1}\right]\left[\frac{2x}{x^2+1} + \frac{1}{2}\cdot\frac{1}{x+3} - \frac{1}{x-1}\right]$

2^π : $\pi = 3.14\ldots$ $\quad 2^\pi$ = limit of $\{2^3, 2^{\frac{31}{10}}, \ldots\}$

$2^\pi = e^{\pi \cdot \ln 2}$ $\quad$ define $x^r = e^{\ln x^r} = e^{r \cdot \ln x}$, $x > 0$

$$\frac{d}{dx} x^r = r \cdot x^{r-1} = \frac{d}{dx} e^{r \cdot \ln x} = e^{r \cdot \ln x} \cdot r \cdot \frac{1}{x}$$

$$= x^r \cdot r \cdot \frac{1}{x} = r \cdot x^{r-1}$$

ex.

$\frac{d}{dx} x^x = ?$ $\quad y = x^x \Longleftrightarrow \ln y = \ln x^x = x \cdot \ln x$ $\quad$ 2nd way

$$\frac{1}{y} \cdot y' = 1 \cdot \ln x + x \cdot \frac{1}{x} \qquad \frac{d}{dx} e^{x \cdot \ln x}$$

$$y' = x^x \cdot (\ln x + 1)$$

Theorem: $e = \lim_{x \to 0} (1+x)^{1/x}$

Proof: $f(x) = \ln x$ $\quad 1 = f'(1) = \lim_{h \to 0} \frac{\ln(1+h) - \ln 1}{h}$

$$1 = \lim_{h \to 0} \frac{1}{h} \ln(h+1) = \lim_{x \to 0} \ln(1+x)^{1/x}$$

$$1 = \ln \underbrace{\left(\lim_{x \to 0} (1+x)^{1/x} \right)}_{\text{must be } e}$$

Alternative definitions

$e = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x$ $\quad e \approx \left(1 + \frac{1}{100}\right)^{100}$

ex.

$$y=(\ln x)^{\ln x} \qquad \ln y = \ln(\ln x)^{\ln x} = \ln x \cdot \ln(\ln x)$$

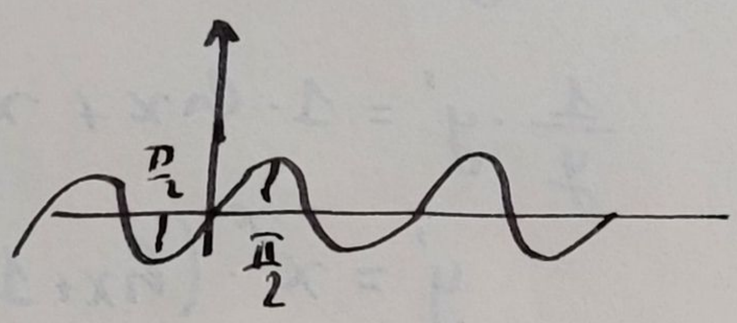
$$\frac{1}{y}\cdot y' = \frac{1}{x}\cdot \ln(\ln x) + \ln x \cdot \frac{1}{\ln x}\cdot\frac{1}{x}$$

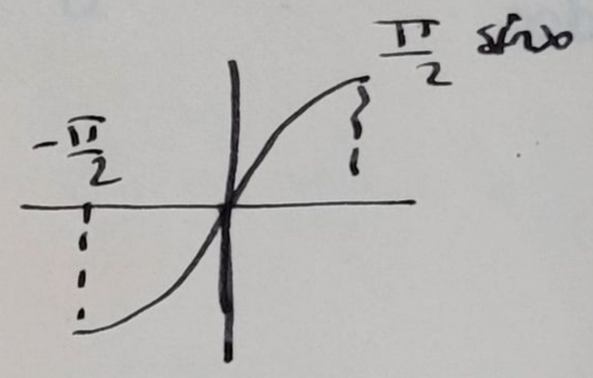
$$y' = (\ln x)^{\ln x}\cdot\left[\frac{1}{x}\cdot\frac{\ln x}{\ln x} + \frac{1}{x}\cdot\ln(\ln x)\right]$$

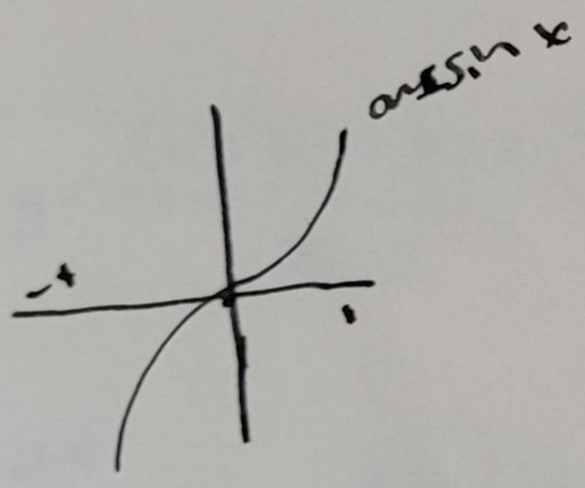
$$y' = (\ln x)^{\ln x}\cdot\left[\frac{1}{x}\ln(\ln x) + \frac{1}{x}\right]$$

3.9 Inverse Trigonometric Functions

$y = \sin^{-1}(x) = \arcsin x$



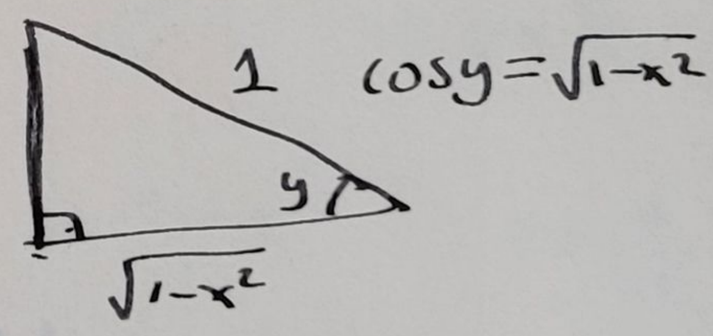




$\frac{d}{dx}\sin^{-1}x \qquad y = \sin^{-1}x \qquad \frac{d}{dx}\sin y = \frac{d}{dx}x = 1$

$\sin y = x$

$$\frac{d}{dx}\sin y = \cos y\cdot\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$



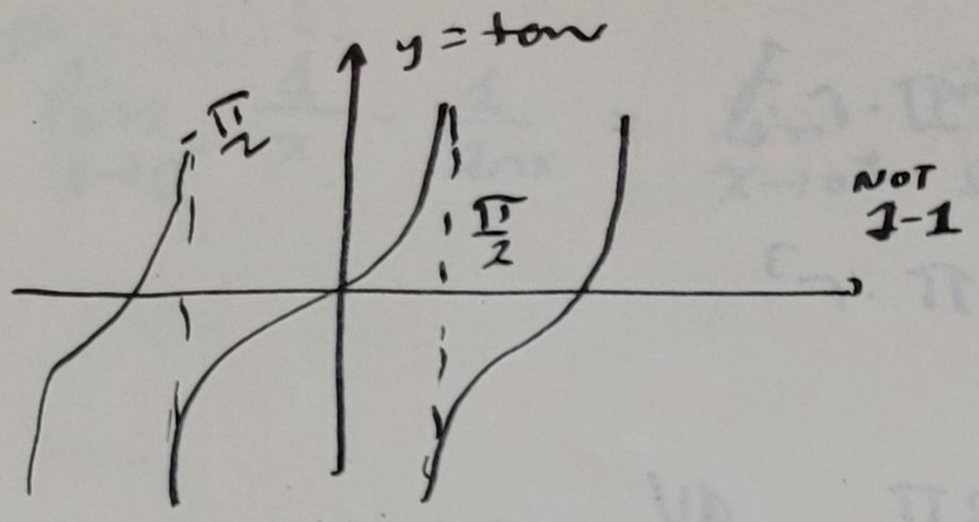
$\cos y = \sqrt{1-x^2}$

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}, \quad -1<x<1$$

$1-\sin^2$

$$\frac{d}{dx}\sin^{-1}u(x) = \frac{1}{\sqrt{1-u^2(x)}}\cdot u'(x)$$

ex. $\frac{d}{dx}\sin^{-1}x^2 = \frac{1}{\sqrt{1-x^2}}\cdot 2x$



$\tan y = x \Leftrightarrow y = \tan^{-1} x, \quad -\infty < x < \infty$

$-\frac{\pi}{2} < y < \frac{\pi}{2}$

$x = -\pi/2$, $x = \pi/2$ } vertical asymptotes of $y = \tan x$

$y = -\pi/2$, $y = \pi/2$ } horizontal asymptotes of $y = \tan^{-1} x$

$y = \tan^{-1} x \Leftrightarrow x = \tan y$

$$1 = \frac{dy}{dx} \cdot \tan y = (1 + \tan^2 y) \cdot \frac{dy}{dx} = (1 + x^2) \cdot \frac{dy}{dx}$$

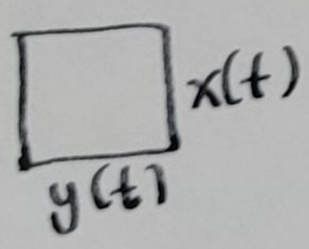
$$\boxed{\frac{dy}{dx} = \frac{1}{1+x^2}}$$

ex. $\frac{d}{dx} \tan^{-1}(e^{x^2}) = \frac{1}{1+(e^{x^2})^2} \cdot e^{x^2} \cdot 2x$

$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, \quad |x| < 1$ $\qquad$ $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x| \cdot \sqrt{x^2-1}}, \quad |x| > 1$

3.10 Related Rates

How fast is the area of a rectangle changing if one side is 10cm long and is increasing at a rate of 2cm/s and the other side is 8m long and is increasing at a rate of -3cm/s?



$x(t) = 10cm + t \cdot 2 = 10 + 2t$

$y(t) = 8cm + t \cdot 3 = 8 + 3t$

$\text{Area} = (10+2t)(8+3t)$

$\text{Area} = 80 + 30t + 16t + 6t^2$

$\frac{d}{dt}\text{Area} = 12t + 46$

$$\frac{dA}{dt} = \frac{dx}{dt} \cdot y + x \frac{dy}{dt}$$

$$= 2 \cdot 8 + 10 \cdot (-3)$$

$$= -14 cm^2/s$$

How fast is the surface area of a ball changing when the volume of the ball is $32\pi/3\ cm^3$ and is increasing at the rate $2\ cm^3/s$?

$$V_{Ball} = \frac{4}{3}\cdot\pi\cdot r^3$$

$$A_{Ball} = 4\cdot\pi\cdot r^3$$

$$V = \frac{32\pi}{3} \qquad \frac{dV}{dt} = 2$$

$$\frac{dA}{dt} = 4\pi\cdot 2r\cdot\frac{dr}{dt}$$

$$\frac{8}{\frac{32\pi}{3}} = \frac{4}{3}\cdot\pi\cdot r^3 \Rightarrow r = 2\ cm$$

$$\frac{dV}{dt} = 4\cdot\pi\cdot r^2\cdot\frac{dr}{dt} = 2 \rightarrow \frac{dr}{dt} = \frac{1}{8\pi}$$

Indeterminate Forms

There are other types of indeterminate forms

$$\left[\frac{0}{0}\right],\ \left[\frac{0}{\infty}\right],\ \left[\frac{\infty}{\infty}\right],\ \left[\frac{-\infty}{\infty}\right],\ [0^\infty],\ [1^\infty],\ [\infty^\infty]$$

$$\lim_{x\to 0}\frac{x}{x} = 1 \qquad \lim_{x\to 0}\frac{x}{x^2} = +\infty \qquad \lim_{x\to 0}\frac{x^2}{x} = 0$$

Theorem: L'Hospital

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} g(x) = 0 \quad (\text{or } \infty, \text{ or } -\infty)$$

If $\lim_{x\to a}\frac{f(x)}{g(x)}$ exist then $\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$

ex.

$$\lim_{x\to 1}\frac{\ln x}{x^2-1} = \lim_{x\to 1}\frac{1/x}{2x} = \frac{1}{2}$$

ex

$$\lim_{x\to 0}\frac{2\sin x - \sin 2x}{2e^x - 2 - 2x - x^2} \Rightarrow \frac{2\cos x - 2\cos 2x}{2e^x - 2 - 2x} \Rightarrow \frac{-2\sin x + 4\sin 2x}{2e^x - 2}$$

$$\Rightarrow \lim_{x\to 0}\frac{-2\cos x + 8\cos 2x}{2e^0} = 3$$

ex.

$$\lim_{x\to 1^+}\frac{x}{\ln x} = +\infty$$

ex.

$$\lim_{x\to 0^+} \frac{1}{x} - \frac{1}{\sin x} = \lim_{x\to 0^+} \frac{\sin x - x}{\sin x \cdot x} \left[\frac{0}{0}\right] \Rightarrow \lim_{x\to 0^+} \frac{\cos x - 1}{x\cdot\cos x + \sin x}$$

$$= \lim_{x\to 0^+} \frac{-\sin x}{\cos x + \cos x - x\sin x} = \frac{0}{2} = 0$$

ex

$$\lim_{x\to 0^+} x^x = \lim_{x\to 0^+} e^{\ln x^x} = \lim_{x\to 0^+} \exp(\ln x^x) = \exp\left(\lim_{x\to 0^+} x\cdot \ln x\right)$$

$[0^0]$

$$= \exp\left(\lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x}}\right)$$

$$= \exp\left(\lim_{x\to 0^+} \frac{1/x}{(-1/x)}\right)$$

$$= \exp\left(\lim_{x\to 0^+} -x\right) = e^0 = 1$$

$$\lim_{x\to\infty} \left(1+\sin\left(\frac{3}{x}\right)\right)^x \quad [1^\infty] = \lim_{x\to\infty} \exp\left(\ln\left(1+\sin\left(\frac{3}{x}\right)^x\right)\right)$$

$$= \exp\left(\lim_{x\to\infty} x\cdot \ln\left(1+\sin\frac{3}{x}\right)\right) [\infty\cdot 0]$$

$$= \exp\left(\lim_{x\to\infty} \frac{\ln\left(1+\sin\frac{3}{x}\right)}{1/x}\right) = \exp\left(\lim_{x\to\infty} \frac{\frac{1}{1+\sin\frac{3}{x}}\cdot\cos\frac{3}{x}\cdot\frac{-3}{x^2}}{-1/x^2}\right)$$

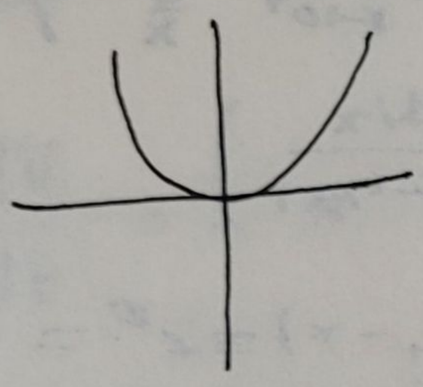
$$= \exp\left(\frac{1}{1+\sin 0}\cdot\cos 0\cdot 3\right) = e^3$$

4.1 Extreme Values of Functions

$y = f(x)$, $x \in D$ D: domain

If $f(x) \le f(x_0)$ for all $x \in D$ then $f(x) \ge f(x_0)$

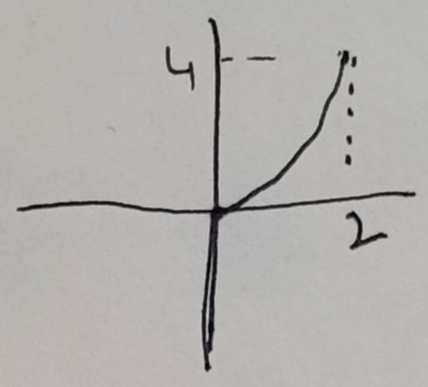
$f(x_0)$ = absolute maximum of f on D. minimum of f on D



$y = x^2$, $x \in R$

No absolute max

0 is abs. min



$y = x^2$, $x \in [0,2]$

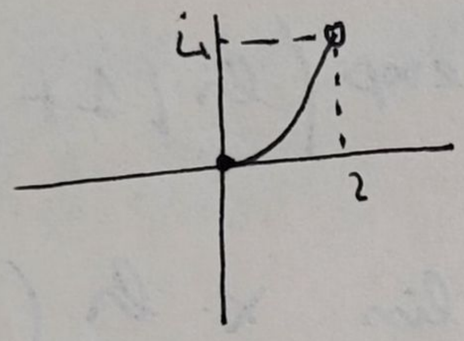
4 = abs. max

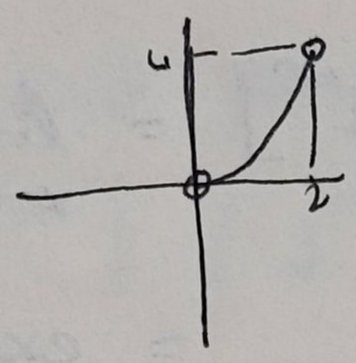
0 = abs. min

$y = x^2$, $x \in [0,2)$

No abs. max

0 abs. min



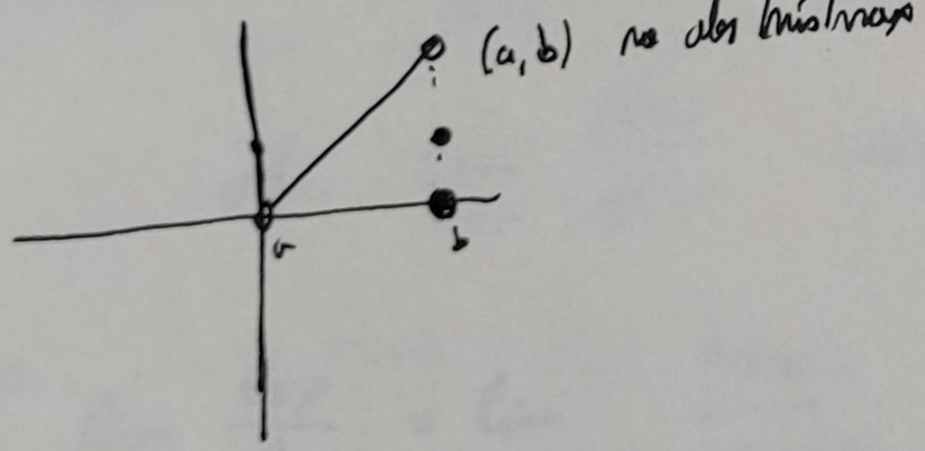


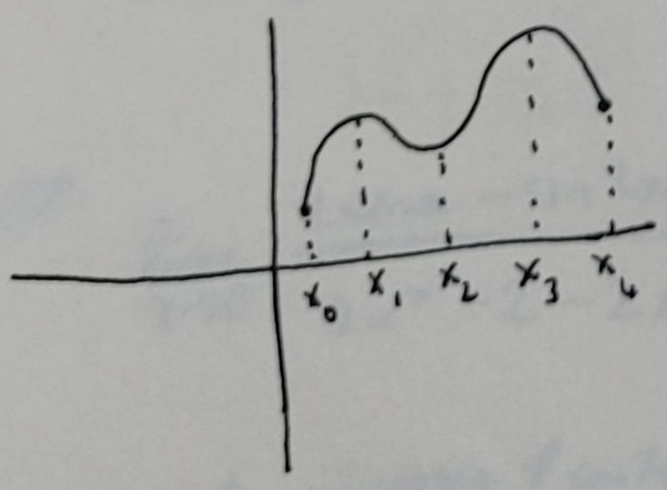
$y = x^2$, $x \in (0,2)$

no abs. max/min

- The Extreme Value Theorem

If $Dom(f) = [a,b] \Rightarrow f$ attains abs max/min values on $[a,b]$ and f is continuous on $[a,b]$





x_0: no abs min

x_3: abs max

x_1, x_3: local max

x_0, x_2, x_4: local min

THEOREM

If f has a local max or min at an interior point c of its domain and $f'(c)$ is defined then $f'(c) = 0$

proof:

suppose $f(c)$ local max

$$f'(c) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c} = \lim_{x\to c^+} \underbrace{\frac{f(x)-f(c)}{x-c}}_{\geq 0} \leq 0 = \lim_{x\to c^-} \underbrace{\frac{f(x)-f(c)}{x-c}}_{\leq 0} \geq 0$$

$f'(c) \leq 0$ and

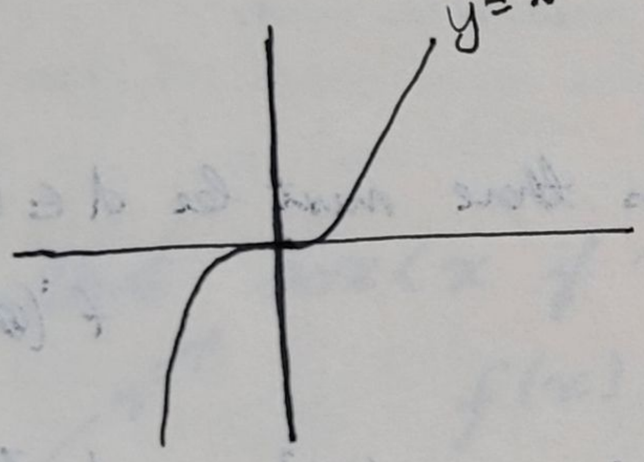
$f'(c) \geq 0$

$f'(c) = 0$

Local max/min can only occur at

1) interior points where $f' = 0$
2) interior points where f' is undefined } critical points of f
3) end points of the domain

ex. • $y = x^3$



$y' = 3x^2 = 0$

but $f(0)$ isn't max/min

ex • Find the abs. max and min of $f(x) = 10x(2-\ln x)$, $x \in [1, e^2]$

$$f'(x) = 10\cdot(2-\ln x) + 10x\cdot\left(\frac{-1}{x}\right) = 10(1-\ln x)$$

1) $f' = 0 \Rightarrow 10(1-\ln x) = 0 \Rightarrow x = e$ 2) f' und. $f'(x)$ defined $[1, e^2]$

3) end points $1, e^2$ $f(1) = 20$ $f(e) = 10e$ $f(e^2) = 0$

abs max abs min

• $f(x) = 3x^{2/3} - 2x$, $x \in [-1, 1]$

$$f'(x) = 3\cdot\frac{2}{3}\cdot x^{-1/3} - 2$$

$$f'(x) = \frac{2}{\sqrt[3]{x}} - 2$$

1) $f'(x) = 0 \Rightarrow$ (x = 1)

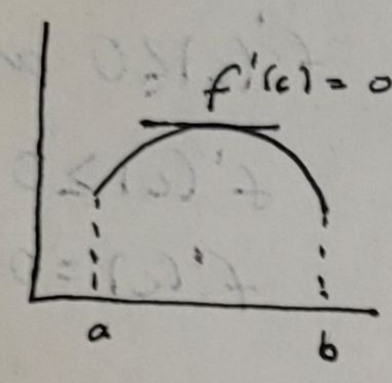
2) $f'(x)$ is undefined at (x = 0) abs min

3) End points (x = -1), (x = 1)

abs max

4.2 MEAN VALUE THEOREM

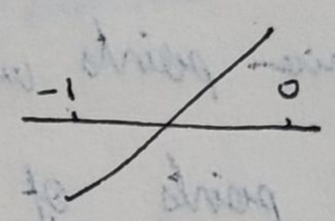
$y=f(x)$ cts on $[a,b]$ and diff on (a,b) If $f(a)=f(b)$ then there is $c\in(a,b)$, $f'(c)=0$



proof: $f(x)$ is cts on $[a,b]\Rightarrow$ extreme value theorem holds $\Rightarrow$ f has abs min/max on $[a,b]$

- Show that has exactly one solution:

$x^3+3x+1=0$ ✱ $f(0)=1>0$ $f(-1)=-3<0$



f is cts. By intermediate value theorem, there is $c\in(-1,0)$ $f(c)=0$

Suppose there are two solutions:

$f(c_1)=f(c_2)=0$ By Rolle's theorem there must be $d\in(c_1,c_2)$

$f'(d)=0$

$y=f(x)$ cts on $[a,b]$ diff on (a,b) then there is $c\in(a,b)$ such that

$$\frac{f(b)-f(a)}{b-a}=f'(c)$$

COROLLARY If $f'(x)=0$ at each point $x\in(a,b)$ then $f(x)=C$ for all $x\in(a,b)$

If $f'(x)=g'(x)$ at each $x\in(a,b)$ then $f(x)=g(x)+C$

- It took 14 sec for a mercury thermometer to rise from $-19°C$ to $100°C$ show that somewhere along the way the mercury was rising at the rate $8.5°C/sec$

$f(t)$ = temp at time t is $°C$

$f(0) = -19$

$f(14) = 100$

By MVT $\dfrac{f(14)-f(0)}{14-0} = f'(c) = \dfrac{100+19}{14} \approx 8.5$

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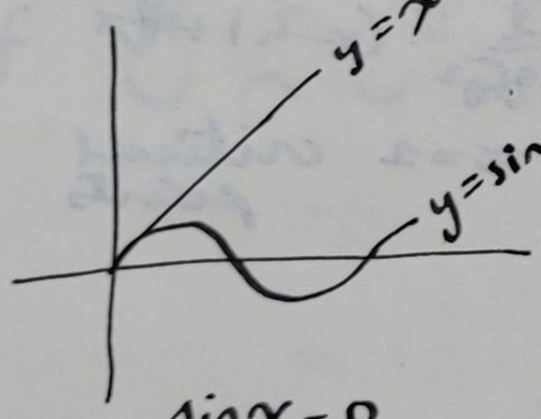
- $f(x) = |x|, \quad x \in [-1,1]$

$\dfrac{f(1)-f(-1)}{1-(-1)} = \dfrac{|1|-|-1|}{1+1} = 0$

But there is no $c \in (-1,1]$

$f'(c) = 0$ MVT does not hold here because the function is not differentiable $x=0$

- Show that $\sin x < x$ if $x > 0$



$f(x) = \sin x$ suppose $x > 0$ $\dfrac{f(x)-f(0)}{x-0} = f'(c)$

for some $c \in (0,x)$

$\dfrac{\sin x - 0}{x-0} = \cos c$ for some $c \in (0,x)$ $\sin x = \cos c \cdot x$

If $x \in (0,2\pi] \Rightarrow c \in (0,x) \Rightarrow 0 < c < 2\pi \Rightarrow \cos c < 1$

For $x \in (0,2\pi) \Rightarrow \sin x = x \cdot \cos c < x \quad 1 = x$

For $x \geq 2\pi \Rightarrow x \geq 2\pi > 1 \geq \sin x$

- Show that $\sqrt{1+x} < 1 + \frac{x}{2}$ for all $x > 0$

4.3 MONOTONIC FUNCTIONS AND THE FIRST DERIVATIVE TEST

Suppose f is cts. on $[a,b]$ and diff. on (a,b)

If $f'(x)>0$ for all $x\in(a,b) \Rightarrow f$ is increasing on (a,b)

$f'(x)<0$ for all $x\in(a,b) \Rightarrow f$ is decreasing on (a,b)

- $f(x)=\sqrt{x}$

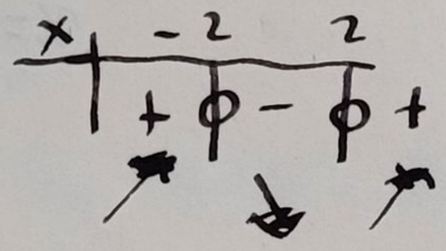
$f'(x)=\frac{1}{2\sqrt{x}}>0$ for all $x>0$

So f is cts on $[0,b]$ and diff. on $(0,b)$ and $f'(x)>0$ for all x is $(0,b]$. So f is increasing on $[0,b]$. This is true for any $b>0$ so the function is increasing on $[0,\infty)$

- $f(x)=x^3-12x-5$

Identify the intervals on which the function is increasing / decreasing.

$f'(x)=3x^2-12 \Rightarrow x^2=4 \Rightarrow x_1=2,\ x_2=-2$



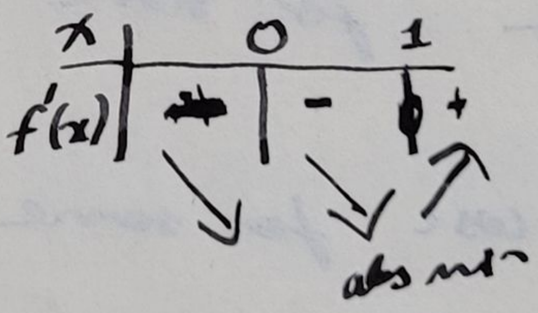
first derivative test:
f has local max at $x=-2$
min at $x=2$

- Find the critical points, identify the open intervals on which f is increasing and decreasing. Find the local and absolute extreme values.

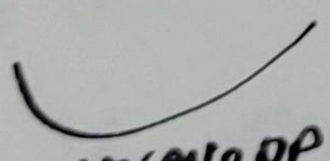
$f(x)=x^{1/3}\cdot(x-4)$

$f'(x)=\frac{1}{3}\cdot x^{-2/3}\cdot(x-4)+x^{1/3}=\frac{1}{3\sqrt[3]{x^2}}\cdot(x-4)+\sqrt[3]{x}$

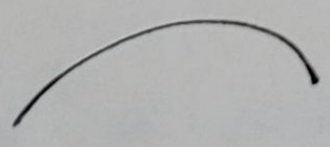
$f'(1)=0$ $f'(0)$ undef. $x=0,\ x=1$ critical points



4.4 CONCAVITY and CURVE SKETCHING



concave up
f' is increasing
$f''>0$



f' is decreasing
$f''<0$

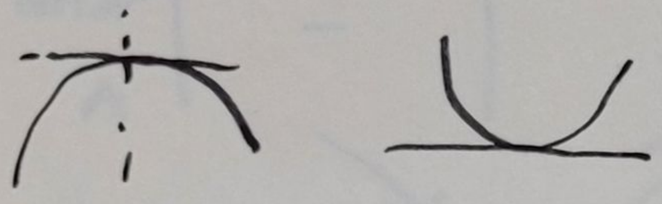
$(c,f(c))$ is called an inflection point if the concavity changes
$f''(c)=0$ or $f''(c)$ undef.

- $y = x^{1/3}$ $\quad x=0$ critical point since y' = undef.

$y' = \frac{1}{3}x^{-2/3} = \frac{1}{3(x^{1/3})^2}$ $\quad y'' = \frac{-2}{9}x^{-5/3} = \frac{-2}{9(x^{1/3})^5}$

		0	
y'	+	$\exists$	+
y''	+	|	−
	∩	inf	∪

SECOND DERIVATIVE TEST for LOCAL EXTREME



$f'(x)=0$, $f''(x)<0$ } $f(x)$ is a local max

$f'(x)=0$, $f''(x)>0$ } $f(x)$ is a local min

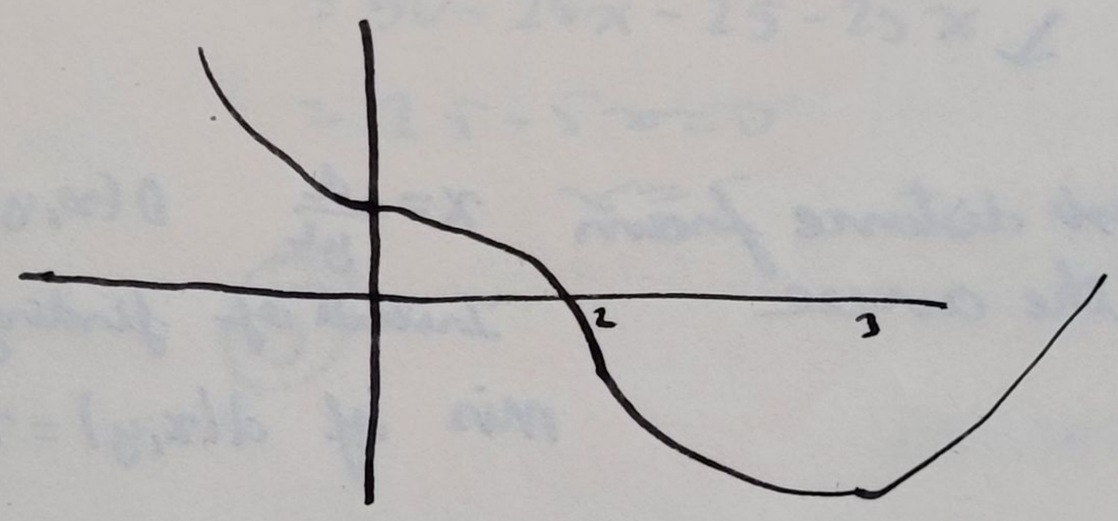
- Sketch the graph of $f(x) = x^4 - 4x^3 + 10$

$f'(x) = 4x^3 - 12x^2 + 10 = 4x^2(x-3) \Rightarrow x=0,\ x=3$ critic. pair.

$f''(x) = 12x^2 - 24x = 12x(x-2) \Rightarrow x=0,\ x=2$

x		0		2		3	
f'	−	0	−		−	0	+
f''	+	0	−	0	+		+
f	↘	→	↘	→	↘	→	↗
	∪	∩	∪	∩	∪	∩	∪

$\lim_{x\to\mp\infty} f(x) = +\infty$ $\quad f(0)=10$ $\quad f(2)=-6$ $\quad f(3)=-17$



- Sketch the graph of $y = \frac{(x+1)^2}{1+x^2} \geq 0$

$\lim_{x\to\mp\infty} \frac{(x+1)^2}{(1+x^2)} = 1$

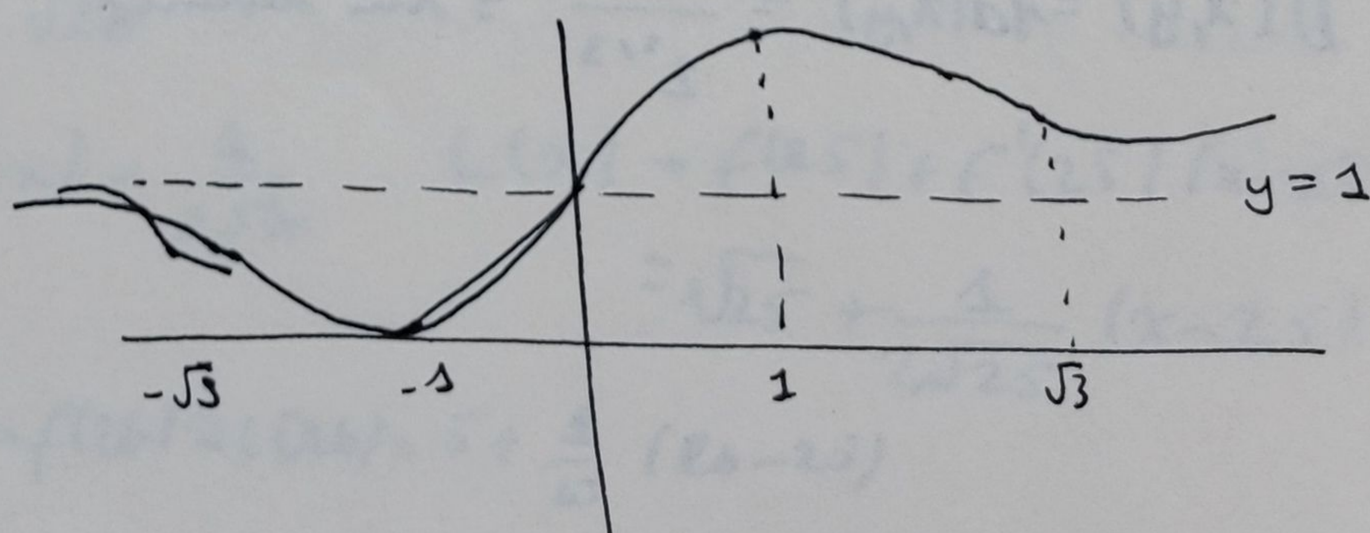
① Domain = $\mathbb{R}$

② No vertical asym.
$y=1$ horizontal asym.

③ $y' = \frac{2(1-x^2)}{(1+x^2)^2}$ $\quad$ ④ $y'=0 \Rightarrow x = \mp 1$

⑤ $y'' = \frac{4x(x^2-3)}{(1+x^2)^3}$ $\quad y''=0 \Rightarrow x=0,\ x=\mp\sqrt{3}$

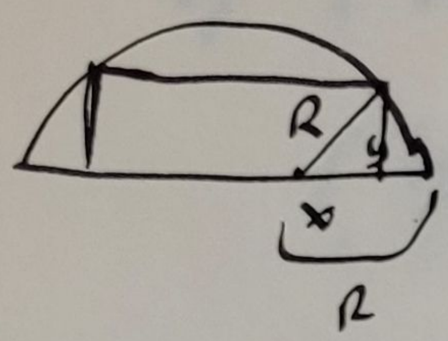
x		$-\sqrt{3}$		-1		0		1		$\sqrt{3}$	
y'	−		−	0	+		+	0	−		−
y''	−	0	+		+	0	−		−	0	+
y	↘	→	↘	→	↗	→	↗	→	↘	→	↘
	∩	inf	∪	max	∪	inf	∩	min	∩	inf	∪



4.6 APPLIED OPTIMIZATION

02/12/2019

Find the area of the largest rectangle that can be drawn inside a semicircle of radius R as:



$A = 2xy$ $\quad y = +\sqrt{R^2 - x^2}$

$x^2 + y^2 = R^2$

~~$A = f(x) =$~~

$A(x) = 2x\sqrt{R^2 - x^2}, \quad 0 < x < R$

$$\frac{dA}{dx} = 2\sqrt{R^2-x^2} + 2x \cdot \frac{-1}{2\sqrt{R^2-x^2}} = \frac{2(R^2-x^2) - 2x^2}{\sqrt{R^2-x^2}} = \frac{2R^2 - 4x^2}{\sqrt{R^2-x^2}}$$

$$\frac{dA}{dx} = 0 \iff 2R^2 - 4x^2 = 0 \qquad x^2 = \frac{R^2}{2} \qquad x = \cancel{-\frac{R}{\sqrt{2}}}, \ \boxed{x = \frac{R}{\sqrt{2}}}$$

x	0		$R/\sqrt{2}$		R
$\frac{dA}{dx}$	|	+	|	-	|
A		↗		↘	

$$A\left(\frac{R}{\sqrt{2}}\right) = R^2$$

- Find the shortest distance from the origin to the curve $x^2 \cdot y^4 = 1$

$x = \frac{1}{y^2}$ $\quad D(x,y) = \sqrt{(x-0)^2 + (y-0)^2}$

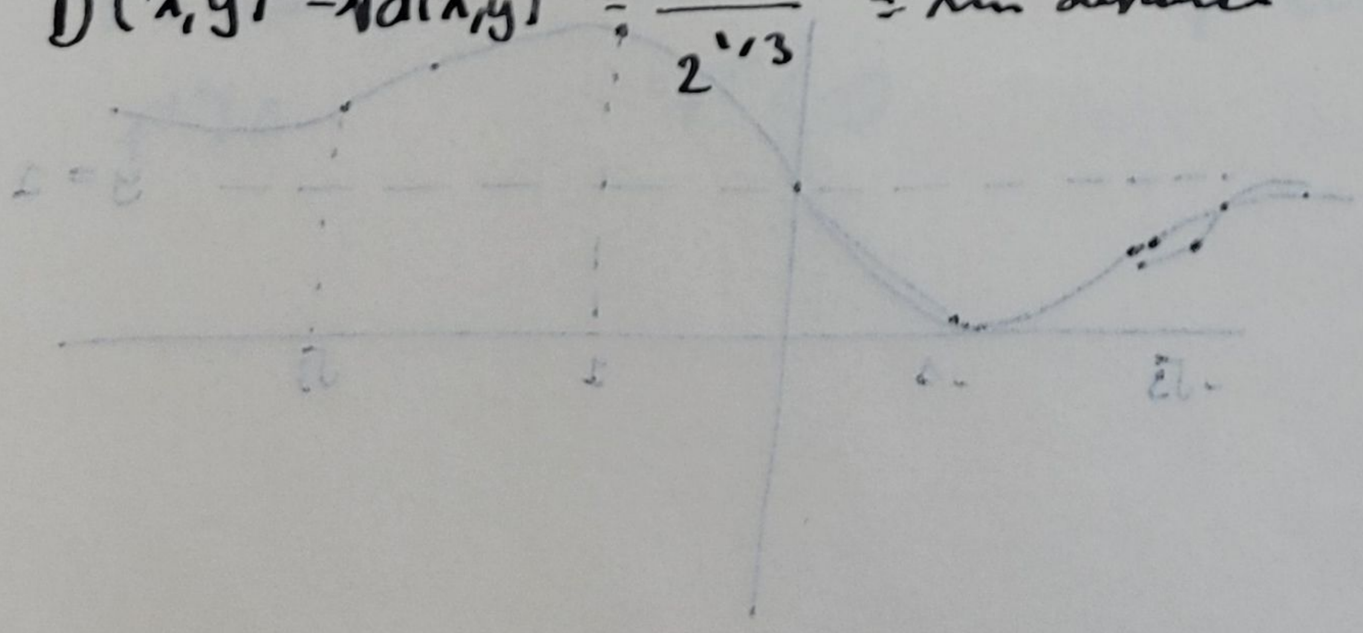
Instead of finding min of D find the min of $d(x,y) = x^2 + y^2$ $\quad x^2 = \frac{1}{y^4}$ ➡

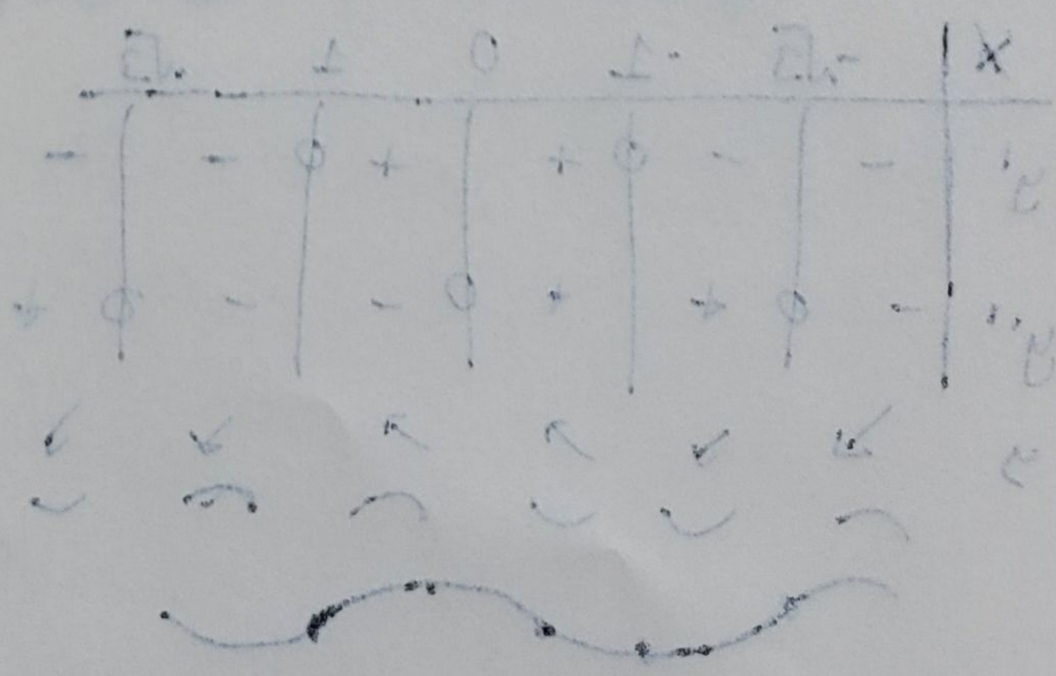
$$d(y) = \frac{1}{y^4} + y^2$$

$$d'(y) = \frac{-4}{y^5} + 2y \qquad d'(y) = 0 \iff \frac{-4 + 2y^6}{y^5} = 0$$

$y^6 = 2$ $\quad y = \pm 2^{1/6} \Rightarrow x = \pm \frac{1}{2^{1/3}}$ $\quad d(x,y) = x^2 + y^2 = \frac{1}{2^{2/3}} + 2^{1/3} = \frac{3}{2^{2/3}}$

$$D(x,y) = \sqrt{d(x,y)} = \frac{3^{1/2}}{2^{1/3}} = \text{min distance}$$





• Design a 1L can shaped right circular cylinder that uses the least material.

$V = \pi r^2 h = 1 \qquad h = \frac{1}{\pi r^2}$

$\text{Area} = 2\pi r^2 + 2\pi rh$

$A = 2\pi r^2 + 2\pi r \cdot \frac{1}{\pi r^2} = 2\pi r^2 + \frac{2}{r}$

$\frac{dA}{dr} = 4\pi r - \frac{2}{r^2} = 0 \qquad r^3 = \frac{2}{4\pi} \qquad r = \sqrt[3]{\frac{1}{2\pi}}$

r		$\sqrt[3]{1/2\pi}$	
$\frac{dA}{dr}$	−	|	+
A	↘		↗

$A = 2\pi \left(\frac{1}{2\pi}\right)^{2/3} + \frac{2}{\left(\frac{1}{2\pi}\right)^{1/3}} = \min$

• You have 100 tons of metal. Sell now with profit \$5/ton. Each week you produce 10 tons of metal. Each ~~metal~~ week price drops by \$0.25/ton. Best time for selling?

x = weeks $\qquad$ Metal $(x) = 100 + 10x$

Profit $(x) = 5 - 0.25x$

T = Total profit $= (100+10x)(5-0.26x), \quad x \geq 0$

$\frac{dT}{dx} = 10(5-0.25x) + (100+10x)\cdot(-0.25)$

$= 50 - 25x - 25 - 25x$

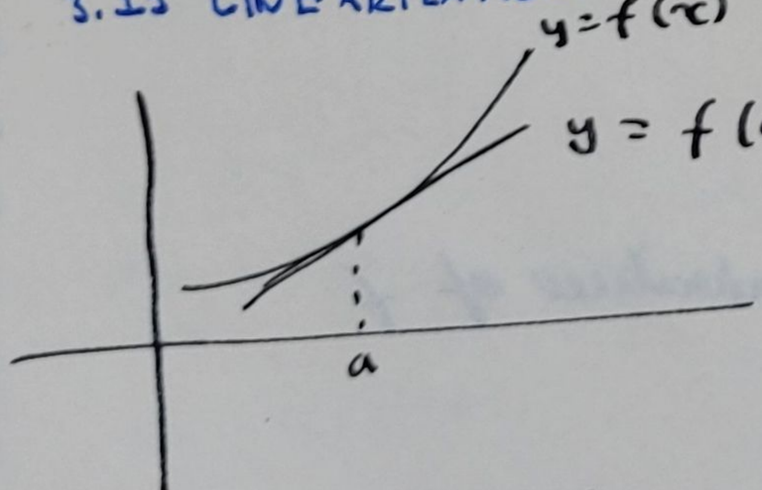
$= 25 - 5x = 0$

$x = 5$

x		0		5	
$\frac{dT}{dx}$		|	+	|	−
T		↗		↘	

(5)

3.11 LINEARIZATION



$y = f(a) + f'(a)(x-a) = L(x) \approx f(x)$ when x is near a

• Using linearization, approximate $\sqrt{26}$

$f(x) = \sqrt{x} \qquad a = 25 \qquad f'(x) = \frac{1}{2\sqrt{x}}$

$L(x) = f(25) + f'(25)(x-25)$

$= \sqrt{25} + \frac{1}{2\sqrt{25}}(x-25)$

$L(x) = 5 + \frac{1}{10}(x-25) \qquad \sqrt{26} = f(26) \approx L(26) = 5 + \frac{1}{10}(26-25)$

$\downarrow$

$5,099 \approx 5,1$

• Approximate $\cos\frac{\pi}{5} = \cos 36^\circ$ using the linearization at $x = \frac{\pi}{6}$

$f(x) = \cos x$ $\qquad L(x) \simeq \frac{\sqrt{3}}{2} - \frac{1}{2}\left(x - \frac{\pi}{6}\right)$

$f'(x) = -\sin x$

$\cos\frac{\pi}{5} \approx L\left(\frac{\pi}{5}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2}\cdot\frac{\pi}{30}$

$L(x) = f(x) + f'(x)(x - \frac{\pi}{6})$

$L(x) = \cos\frac{\pi}{5} - \sin\frac{\pi}{5}\left(\frac{\pi}{30}\right)(x - \frac{\pi}{6})$

Error Estimation

The error in the approximation is $\frac{f''(s)}{2}(x-a)^2$ where s is a number between a and x.

• Give an upper bound for the error in the last problem

$f(x) = \sqrt{x}$ $\quad f'(x) = \frac{1}{2\sqrt{x}}$ $\quad f''(x) = \frac{-1}{4x^{3/2}}$ $\quad s$ is between $a=25$ and $x=26$

$|f''(s)| = \left|\frac{-1}{4s^{3/2}}\right| = \frac{1}{4}\cdot\left|\frac{1}{\sqrt{s^3}}\right| \leq \frac{1}{4}\cdot\frac{1}{25^{3/2}} = \frac{1}{500}$

$|\text{Error}| = \frac{1}{2}\cdot|f''(s)|(x-a)^2 \leq \frac{1}{2}\cdot\frac{1}{500}\cdot 1^2 = \frac{1}{1000}$

$5.099 = 5.1 - \frac{1}{1000} < \sqrt{26} < 5.1 + \frac{1}{1000} = 5.101$

4.8 ANTI DERIVATIVES 3/11/19

F is called an anti-derivative of f on an interval I if

$F'(x) = f(x)$ for all $x \in I$

$f(x) = x^2$ $\quad F(x) = \frac{x^3}{3}$

$F(x) = \frac{x^3}{3} + 17$ } both are anti-derivatives of f

THEOREM If $F' = f \Rightarrow$ the most general anti derivative of $f(x)$ is $F(x) + C$

$F' = f$ $\quad (kF)' = kf$ $\quad G' = g$ $\quad (f+G)' = g + f$

k cons.

Indefinite Integrals

The collection of all anti derivatives of f is called indefinite integral of f and is shown as $\int f(x)\,dx$ $\quad \int x^2\,dx = \frac{x^3}{3} + C$

CHAPTER 5 - INTEGRALS

If f is a cts. on $[a,b]$ then f is integrable on $[a,b]$

09/12/2019

5.2 FUNDAMENTALS OF CALCULUS
THEOREM

THEOREM:

pwd I Suppose f is cts on $[a,b]$ then for $x \in (a,b)$

$$\frac{d}{dx}\int_a^x f(s)\cdot ds = f(x)$$

$$F(x) = \int_a^x f(s)\cdot ds \qquad \frac{d}{dx}F(x) = \lim_{h\to 0}\frac{F(x+h)-F(x)}{h}$$

$$= \lim_{h\to 0}\frac{1}{h}\left(\int_a^{x+h} f(s)ds - \int_a^x f(s)ds\right)$$

$$= \lim_{h\to 0}\frac{1}{h}\left(\int_a^x f(s)ds + \int_x^{x+h} f(s)ds - \int_a^x f(s)ds\right)$$

$$= \lim_{h\to 0}\frac{1}{h}\int_x^{x+h} f(s)ds$$

- $\frac{d}{dx}\int_3^x \sin(t^3)\cdot dt = \sin x^3$
- $\frac{d}{dx}\int_x^3 e^{-t^2} = \frac{d}{dx}\cdot -1\cdot\int_3^x e^{-t^2} = -e^{-x^2}$
- $\frac{d}{dx}\int_{3000}^{3x} e^{-t^2}dt = \frac{d}{dx}F(3x) = 3\cdot F'(3x) = 3e^{-(3t)^2}$

$$F(x) = \int_{3000}^x e^{-t^2}dt \Rightarrow F'(x) = e^{-x^2}$$

- $\frac{d}{dx}\int_{x^2}^{x^3} t^5\cdot e^{-t^2}dt$

* $\frac{d}{dx}\int_{f(x)}^{g(x)} H(s)ds = H(g(x))\cdot g'(x) - H(f(x))\cdot f'(x)$

PART II

$F(x) = \int f(x)\,dx$ means $F'(x) = f(x)$

then $\int_a^b f(x)\,dx = F(b) - F(a) = \int_a^b F'(x)\,dx$

proof:

$G(x) = \int_a^x F'(s)\,ds$ $\quad \frac{d}{dx} G(x) = G'(x) = F'(x)$ by FTC I

$\Updownarrow$ mean value theorem

$G(x) = F(x) + C$

$0 = \int_a^a F'(s)\,ds = G(a) = F(a) + C \rightarrow C = -F(a)$

$F(b) + C = G(b) = \int_a^b F'(s)\,ds$

5.3 METHOD of ~~SUBSTITION~~ SUBSTITION

- $\int (x^3 - 3x^2 + 6x - 9)\,dx = \frac{x^4}{4} - x^3 + 3x^2 - 9x + C$

- $\int \frac{(x+1)^3}{x}\,dx = \int \frac{x^3 + 3x^2 + 3x + 1}{x}\,dx = \frac{x^3}{3} + \frac{3x^2}{2} + 3x + \ln|x| + C$

$\int f'(g(x))\,g'(x)\,dx = f(g(x)) + C$ $\quad \int f'\underbrace{(g(x))}_{u=g(x)} \cdot g'(x)\,dx$ $\quad \frac{du}{dx} = g'(x)$

$\int f'(u)\,du = f(u) + C$ $\quad du = g'(x)\,dx$

- $\int x \cdot \sin(\underbrace{2x^2}_{u})\,dx = \int \frac{\sin(u)\,du}{4} = \frac{-\cos u}{4} = \frac{-\cos 2x^2}{4}$

$u = 2x^2$

$\frac{du}{dx} = 4x$

$du = 4x\,dx$

$x\,dx = \frac{du}{4}$

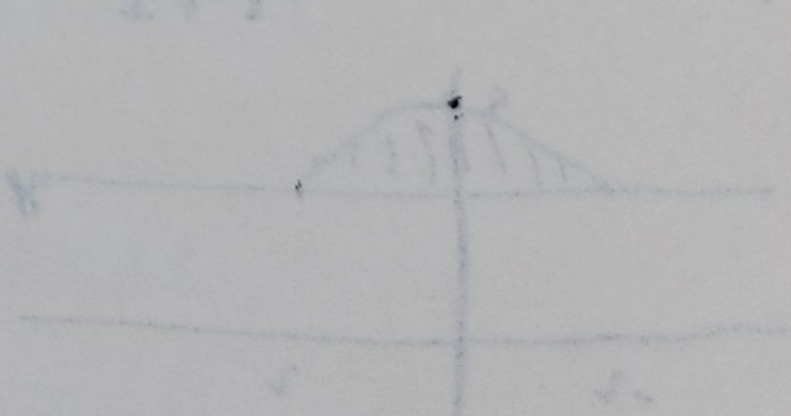
- $\int \sec^2(3x+2)\,dx$

$u = 3x + 2$

$du = 3\,dx$ $\quad \frac{1}{3} \cdot \int \sec^2(u)\,du = \frac{\tan u}{3}$

$dx = du/3$

$\frac{d}{dx} \tan x = \sec^2 x$



- $\int \frac{x}{(x-4)^3} \cdot dx$ $\quad u = x-4 \Rightarrow x = u+4$

$\frac{du}{dx} = 1 \Rightarrow du = dx$

$\int \frac{u+4}{u^3} \cdot du = \int \left(\frac{1}{u^2} + \frac{4}{u^3}\right) \cdot du$

$= -u^{-1} + -2u^{-2} = \frac{-1}{u} - \frac{2}{u^2}$

- $\int \sec x \cdot dx = \int \sec x \cdot \frac{(\sec x + \tan x)}{\sec x + \tan x} \cdot dx$

$u = \sec x + \tan x$

$\frac{du}{dx} = \sec^2 x + \sec x \cdot \tan x$

$= \int \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x}$

$= \int \frac{du}{u} = \ln|u| + C$

- $\int_e^{e^2} \frac{dt}{t \cdot \ln t} \Rightarrow \int \frac{dt}{t \cdot \ln t} = \int \frac{du}{u} = \ln|u| + C = \ln|\ln t| + C$

↳ $\ln|\ln e^2| - \ln|\ln e| = \ln 2 - \ln 1 = \ln 2$

- $\int \frac{dx}{e^x + 1}$ $\quad u = e^x + 1$ $\quad dx = \frac{du}{e^x} = \frac{du}{u-1}$ $\quad \int \left(\frac{-1}{u} + \frac{1}{u-1}\right) du$

$du = e^x \cdot dx$

$= -\ln|u| + \ln|u-1| + C$

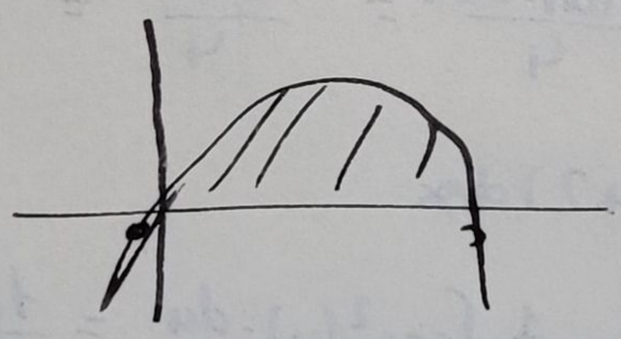
$= -\ln|e^x+1| + \ln|e^x| + C$

10/12/2019

5.4 AREAS OF PLANE REGIONS

- Find the area of the region above the x-axis below $y = 3x - x^2$

$y = 3x - x^2 = x(3-x)$

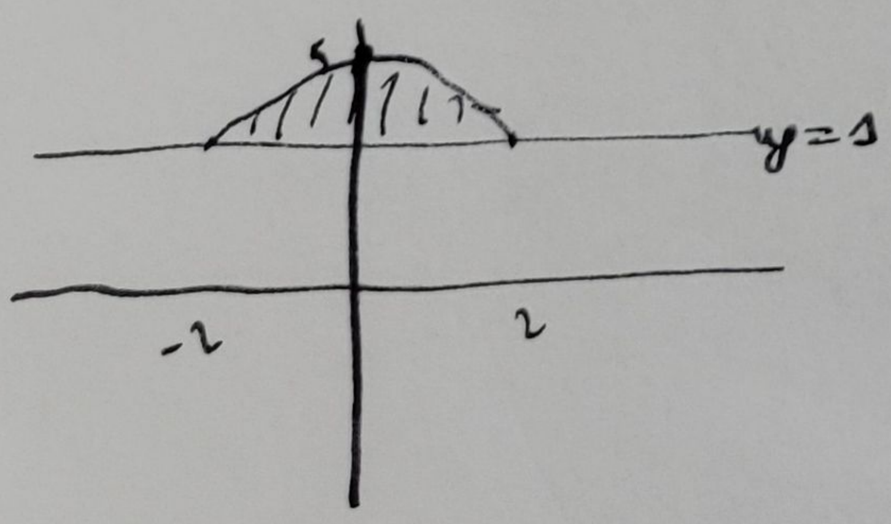


$\int_0^3 3x - x^2 = \frac{27}{2} - \frac{27}{3}$

$\int 3x - x^2 = \frac{3x^2}{2} - \frac{x^3}{3}$

- Find the area above the line $y = 1$ below $y = \frac{5}{x^2+1}$

$2\int_0^2 \frac{5}{x^2+1} - 1 = 10 \arctan 2 - 4$



If

- $f(x) \le g(x)$ for all $x \in [a,b]$ area between the curve $f(x)$ and $g(x)$ between $x=a$ and $x=b$ $\int_a^b (g(x)-f(x))\,dx$

- Find the area between $y=x^2-2x$ and $y=4-x^2$

roots $0, 2$; $-2, 2$

$x^2-2x = 4-x^2$

$2x^2-2x-4=0$

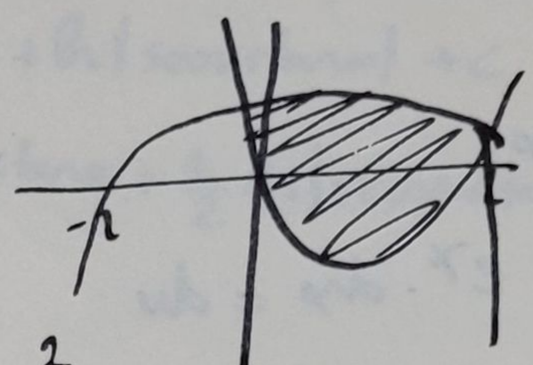
$x^2-x-2=0$

$x_1=-1$

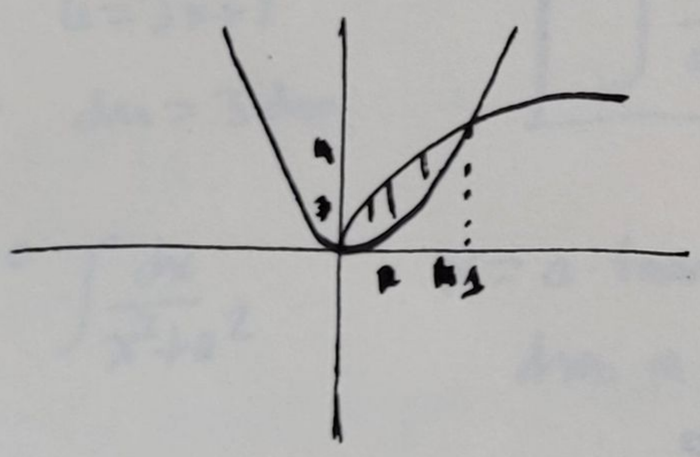
$x_2=2$

$\int_{-1}^{2} (4-x^2)-(x^2-2x)\,dx$

$\int_{-1}^{2} -2x^2+2x+4\,dx$ $\quad -\frac{2x^3}{3}+x^2+4x \Big|_{-1}^{2} = 9$



- Find the area between $y=\sqrt{x}$ and $y=x^2$

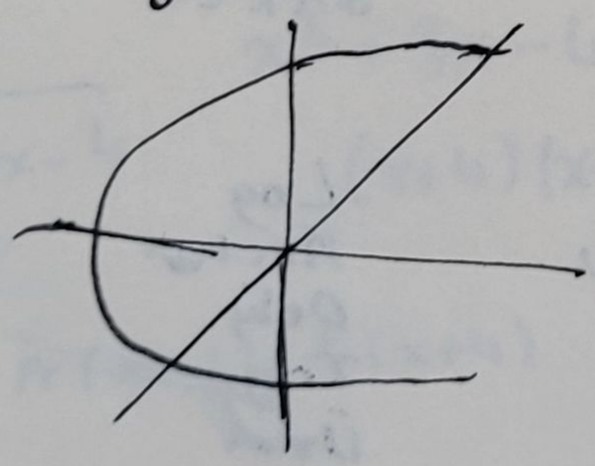


$\int_0^1 \sqrt{x}-x^2 = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{3} \Big|_0^1 = \frac{2}{3} - \frac{1}{3} = \underline{1/3}$

- Find the area between $x=y^2-12$ and $y=x$

$y^2 = x+12$

$y = \sqrt{x+12}$



$\sqrt{x+12} = x$ $\quad x^2 = x+12$ $\quad x^2-x-12=0$

$x \quad -4$

$x \quad +3$ $\quad x_1 = 4 \quad x_2 = -3$

$\int_{-3}^{4} -y^2+12+y\,dy = \frac{343}{6}$

5.5 INTEGRATION BY PARTS

$$\frac{d}{dx}(u\cdot v) = u'v + v'u \qquad d(uv) = du\cdot v + dv\cdot u$$

$$uv = \int d(uv) = \int du\cdot v + \int dv\cdot u \Rightarrow \int u\cdot dv = uv - \int v\cdot du$$

- $\int x\cdot e^x\,dx$

$$\int x\cdot e^x = x\,e^x - \int e^x\,dx = xe^x - e^x + C$$

$u = x \qquad e^x\cdot dx = dv$

$du = dx \qquad e^x = v$

- $\int \ln x\,dx$

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx$$

$$= x\ln x - x + C$$

$u = \ln x \qquad dv = dx$

$du = \frac{1}{x}\cdot dx \qquad v = x$

- $\int x^2 \sin x\,dx$

$u = x^2 \qquad du = 2x\,dx \qquad -x^2\cos x - \int(-\cos x)\,2x\,dx$

$\int \sin x\,dx = \int dv$

$$-x^2\cos x - 2\underbrace{\int -\cos x\cdot x\,dx}$$

$u = x \qquad \int \cos x\,dx = \int dv$

$du = dx \qquad \sin x = v$

- $I = \int e^x \sin x\,dx$

$$I = \sin x\,e^x - \int e^x\cos x\,dx$$

$u = \sin x \qquad \int e^x\,dx = \int dv$

$du = \cos x\,dx \qquad e^x = v$

$u = \cos x \qquad e^x\,dx = dv$

$du = -\sin x\,dx \qquad e^x = v$

Log
Arctan
Poly
Trig
Üstel

~~$I = \sin x\cdot e^x$~~ $\quad I = \sin x\cdot e^x - \left(\cos x\cdot e^x - \int e^x\cdot(-\sin x)\,dx\right)$

$$I = \sin x\,e^x - \cos x\,e^x - \underbrace{\int e^x \sin x\,dx}_{I}$$

$$2I = \sin x\cdot e^x - \cos x\,e^x$$

$$I = \frac{(\sin x - \cos x)e^x}{2} + C$$

• $\int \sec^3 x \, dx = I$ $\quad u = \sec x$ $\quad \int dv = \int \sec^2 x \, dx$

$du = \sec x \cdot \tan x$ $\quad v = \tan x$

$I = \sec x \cdot \tan x - \int \tan x \sec x \tan x \, dx$

$\tan^2 x + 1 = \sec^2 x$

$\tan^2 x = \sec^2 x - 1$

$I = \sec x \cdot \tan x - \int \sec x \cdot (\sec^2 x - 1) \, dx$

$= \sec x \cdot \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$ ($\int \sec^3 x \, dx = I$)

$2I = \sec x \cdot \tan x + \ln|\sec x + \tan x| + C$

$I = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln|\sec x + \tan x| + C$

5.6 INTEGRALS OF RATIONAL FUNCTIONS

$\int \frac{P(x)}{Q(x)} \, dx$ where $P(x)$, $Q(x)$ are polynomials

• $\int \frac{1}{3x+1} \, dx = \int \frac{du/3}{u} = \frac{1}{3} \ln|u| + C = \frac{1}{3} \ln|3x+1| + C$

$u = 3x+1$

$du = 3 \, dx$

$$\int \frac{1}{ax+b} \, dx = \frac{1}{a} \ln|ax+b| + C$$

• $\int \frac{dx}{x^2+a^2}$ $\quad x = a \cdot \tan\theta$ $\quad dx = a \cdot \sec^2\theta \cdot d\theta$

$x^2 + a^2 = a^2 \tan^2\theta + a^2 = a^2 \sec^2\theta$

$$\int \frac{a \sec^2\theta \, d\theta}{a^2 \sec^2\theta} = \frac{1}{a} \int d\theta$$

• $\int \frac{dx}{x^2+3x-4}$ $\quad x^2 + 3x - 4 = 0$ $\quad (x+4)(x-1)$

$\frac{1}{x^2+3x-4} = \frac{1}{(x+4)(x-1)} = \frac{A}{x+4} + \frac{B}{x-1}$

$1 = A(x-1) + B(x+4)$ $\quad x = 1 \Rightarrow B = \frac{1}{5}$ $\quad x = -4 \Rightarrow A = \frac{-1}{5}$

$I = \frac{-1}{5} \int \frac{dx}{x+4} + \frac{1}{5} \int \frac{dx}{x-1} = \frac{-1}{5} \ln|x+4| + \frac{1}{5} \ln|x-1| + C$

$\int \frac{dx}{(x+a)(x+b)}$ $\quad$ ① $a \neq b$ $\quad$ ② $a = b$

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PARTIAL FRACTIONS

- $\int \frac{(x+4)}{x^2-5x+6} \cdot dx = I \quad \frac{x+4}{x^2-5x+6} = \frac{A}{x-2} + \frac{B}{x-3} \quad (x+4) = A(x-3) + B(x-2)$

$x=3 \Rightarrow 7 = B$

$x=2 \Rightarrow 6 = -A$

$I = \int \frac{-6}{x-2} \cdot dx + \int \frac{7}{x-3} dx \quad = -6 \ln|x-2| + 7 \ln|x-3| + C$

5.7 INVERSE SUBSTITUTIONS

if integral contains $\sqrt{a^2-x^2}$ Try $x = a\cdot\sin\theta \Rightarrow \theta = \sin^{-1}\frac{x}{a}$

$$\sqrt{a^2-(a\cdot\sin\theta)^2} = |a|\sqrt{1-\sin^2\theta} = |a||\cos\theta|$$

$\sqrt{a^2-x^2}$ makes sense when $-a \le x \le a$ $\quad \frac{-\pi}{2} \le \theta \le \frac{\pi}{2} \Rightarrow \cos\theta \ge 0$

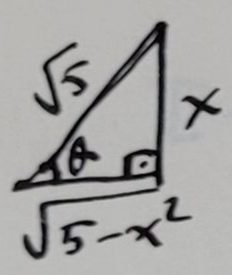
$$\sqrt{a^2-x^2} = |a|\cos\theta$$

- $\int \frac{dx}{(5-x^2)^{3/2}} = \int \frac{dx}{(\sqrt{5-x^2})^3}$ $\quad x = \sqrt{5}\sin\theta$ $\quad \sqrt{5-x^2} = \sqrt{5-5\sin^2\theta}$

$$= \sqrt{5}\sqrt{1-\sin^2\theta}$$
$$= \sqrt{5}\sqrt{\cos^2\theta}$$
$$= \sqrt{5}\cos\theta$$

$dx = \sqrt{5}\cos\theta\, d\theta$

$$I = \int \frac{\sqrt{5}\cos\theta\, d\theta}{(\sqrt{5})^3\cos^3\theta\, d\theta} = \frac{1}{5}\int \frac{1}{\cos^2\theta}\cdot d\theta = \frac{1}{5}\cdot\int \sec^2\theta\, d\theta = \frac{1}{5}\tan\theta + C$$



$\sin\theta = \frac{x}{\sqrt{5}}$ $\quad \tan\theta = \frac{x}{\sqrt{5-x^2}}$ $\quad \int \frac{dx}{(5-x^2)^{3/2}} = \frac{1}{5}\frac{x}{\sqrt{5-x^2}} + C$

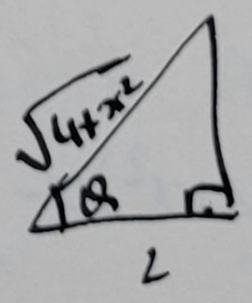
The inverse tangent substitutions

if integral contains $\sqrt{a^2+x^2}$ or $\frac{1}{x^2+a^2}$ $\xrightarrow{\text{Try}}$ $x = a\tan\theta$ or $\theta = \tan^{-1}\frac{x}{a}$ $\quad I = \int \frac{dx}{\sqrt{4+x^2}}$

- $I = \int \frac{dx}{\sqrt{4+x^2}}$ $\quad x = 2\tan\theta$ $\quad dx = 2\sec^2\theta$ $\quad \sqrt{4+x^2} = \sqrt{4+4\tan^2\theta} = 2\sqrt{1+\tan^2\theta} = 2\sqrt{\sec^2\theta} = 2|\sec\theta|$

$$= 2\sec\theta$$

$$I = \int \frac{2\sec^2\theta}{2\sec\theta}\cdot d\theta = \int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$



$\sec\theta = \frac{\sqrt{4+x^2}}{2}$

5.8 EXAMPLES

- $\int \frac{dx}{x(\ln x)^3}$ $\quad \ln x = u$ $\quad du = \frac{1}{x}\cdot dx$ $\quad \int u^{-3}\cdot du = \frac{u^{-2}}{-2} = \frac{1}{-2u^2} \Rightarrow \boxed{\frac{1}{-2\cdot(\ln^2 x)} + C}$

- $\int \frac{x^3}{1+x^8}\,dx$ $\quad \cancel{du = dx}\ \frac{}{x}$ $\quad x^4 = u$ $\quad \int \frac{du/4}{1+u^2} = \frac{1}{4}\arctan u + C$

- $\int \frac{x^3+x+1}{x^2+1}\,dx$ $\quad \begin{array}{r|l} x^3+x+1 & x^2+1 \\ \hline x^3+x & x \\ \hline 1 & \end{array}$ $\quad x + \frac{1}{x^2+1} = \int x\,dx + \int \frac{dx}{x^2+1}$

$= \frac{x^2}{2} + \arctan x + C$

- $\int \arcsin x\,dx = x\arcsin x - \int \frac{x\,dx}{\sqrt{1-x^2}}$ $\quad u = \arcsin x$ $\quad du = \frac{1}{\sqrt{1-x^2}}\cdot dx$ $\quad dw = dx$ $\quad 1-x^2 = w$ $\quad v = x$ $\quad -2x\,dx = dw$

$\int \frac{-dw/2}{\sqrt{w}} = \frac{-1}{2}\cdot\frac{w}{1/2} = -\sqrt{1-x^2}$

- $\int \frac{e^x}{4+e^{2x}}\cdot dx = \int \frac{du}{4+u^2} = 2\int \frac{2\sec^2\theta\,d\theta}{4+4\tan^2\theta} = \int \frac{2\sec^2\theta\,d\theta}{4\sec^2\theta}$

$e^x = u$ $\quad e^x dx = du$ $\quad u = 2\tan\theta$ $\quad du = 2\sec^2\theta\,d\theta$

$= \frac{1}{2}\theta + C$

$= \frac{1}{2}\arctan\left(\frac{u}{2}\right) + C$

$= \frac{1}{2}\arctan\left(\frac{e^x}{2}\right) + C$

- $\int \ln(1+x^2)\,dx$ $\quad u = \ln(1+x^2)$ $\quad du = \frac{1}{1+x^2}\cdot 2x$ $\quad dv = dx$ $\quad x = v$

$= \ln(1+x^2)\,x - \int \frac{x\cdot 2x}{1+x^2}\cdot dx = \ln(1+x^2)\,x - 2(x - \arctan x) + C$

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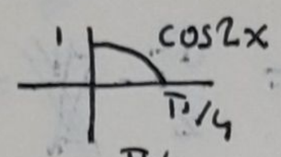
TRIGONOMETRIC INTEGRALS

- eliminating square roots

- $\int_0^{\pi/4} \sqrt{1+\cos 4x}\, dx = I$



$$I = \int_0^{\pi/4} \sqrt{1+\cos^2 2x - 1}\, dx = \int_0^{\pi/4} \sqrt{2}\,|\cos 2x|\, dx = \int_0^{\pi/4} \sqrt{2}\cdot \cos 2x\, dx = \sqrt{2}\cdot \frac{\sin 2x}{2}\Big|_0^{\pi/4}$$

$$= \frac{\sqrt{2}}{2}\cdot\left(\sin\frac{\pi}{2} - \sin 0\right) = \frac{\sqrt{2}}{2}$$

- integrals of powers of tanx and secx

- $\int \tan^4 x\, dx = \int \tan^2 x \tan^2 x\, dx = \int (\sec^2 x - 1)\tan^2 x\, dx = \underbrace{\int \sec^2 x \tan^2 x\, dx}_{I} - \underbrace{\int \tan^2 x\, dx}_{j}$

$u = \tan x \quad du = \sec^2 x$

$$I = \int u^2\, du = \frac{u^3}{3} + C$$

$$j = \int \left[-1 + \overbrace{1 + \tan^2 x}^{\sec^2 x}\right] dx = -x + \int \sec^2 x\, dx = -x + \tan x + C$$

$$I - j = \frac{\tan^3 x}{3} + x - \tan x + C$$

- $\int \sec^3 x\, dx = \int \sec^2 x \cdot \sec x \cdot dx = \int (1+\tan^2 x)\cdot \sec x \cdot dx = I$

$dv = \sec^2 x\, dx \quad v = \tan x$

$u = \sec x \quad du = \sec x \tan x\, dx$

$$I = \int \sec x\, dv + \int \sec x \tan^2 x\, dx$$

$$I = \sec x \cdot \tan x - \int \underbrace{\tan^2 x}_{(\sec^2 x - 1)} \cdot \sec x \cdot dx = \sec x \cdot \tan x - \underbrace{\int \sec^3 x\, dx}_{I} + \underbrace{\int \sec x\, dx}_{\int \frac{\sec x(\sec x + \tan x)\cdot dx}{\sec x + \tan x}}$$

$$2I = \sec x \cdot \tan x + \ln|\sec x + \tan x| + C$$

- $\int \sin^3 x\,dx = \int \sin^2 x \cdot \sin x \cdot dx = \int (1-\cos^2 x)\cdot \sin x \cdot dx = I$

$$I = \int \sin x\,dx - \underbrace{\int \sin x \cdot \cos^2 x\,dx}_{j} = -\cos x + \frac{\cos^3 x}{3} + C$$

$\cos^2 x = u$

$-2\sin x\,dx = du \Rightarrow j = \frac{1}{2}\int u\,du = \frac{u^2}{2}\cdot\frac{1}{2} = \frac{-u^2}{4} + C \Rightarrow \frac{-\cos^3 x}{3} + C$

- $\int \sin^5 x\,dx$

- $\int \sin^3 x \cdot \cos^3 x \cdot dx = \int \sin x (1-\cos^2 x)\cos^3 x\,dx$ $\quad u = \cos x$

$-du = \sin x\,dx$

$$= \int (-du)(1-u^2)u^3 = \int (u^5 - u^3)\,du = \frac{\cos^6 x}{6} + \frac{\cos^4 x}{4} + C$$

- $\int \sec^3 x \cdot \tan x \cdot dx = \underbrace{\int \sec x \cdot \sec^2 x \cdot \tan x\,dx}_{I} = \int \sec x \cdot (1+\tan^2 x)\cdot \tan x\,dx$

$u = \sec x$

$du = \sec x \tan x\,dx$

$$I = \int u^2\,du = \frac{u^3}{3} + C$$

- $\int \sec^4 x \tan^2 x\,dx = \int \sec^2 x \cdot \sec^2 x \cdot \tan^2 x \cdot dx = \int (1+\tan^2 x)\cdot \sec^2 x \cdot \tan^2 x \cdot dx$

$\tan x = u$

$\sec^2 x\,dx = du$

$$\int (1+u^2)\cdot u^2 \cdot du = \int (u^2 + u^4)\,du = \frac{u^3}{3} + \frac{u^5}{5} + C$$

- $\int \sec^4 \theta\,d\theta = \int \sec^2 \theta (1+\tan^2 \theta)\,d\theta$

$\tan \theta = u$

$\sec^2 \theta\,d\theta = du$

$$\int (1+u^2)\,du = u + \frac{u^3}{3} + C$$